Han-Bo-Ram Lee

Professor, Graduate School of Semiconductor Materials & Devices Engineering,
Ulsan National Institute of Science and Technology (UNIST)
hbrlee@unist.ac.kr, https://hbrl-research.group

Executive Editor, Chemistry of Materials, ACS Publications
lee-office@cm.acs.org, http://pubs.acs.org/journal/cmatex

Research Interests

Prof. Lee's current research focuses on bridging AI-driven materials synthesis design with experimental realization by elucidating surface chemical reaction mechanisms for next-generation semiconductor device fabrication. His group actively explores a wide range of theoretical methodologies—including deep learning diffusion models, density functional theory combined with machine-learning force fields, and kinetic Monte Carlo simulations—as well as atomic-scale synthesis techniques such as atomic layer deposition (ALD), atomic layer modulation (ALM), and area-selective deposition (ASD). The overarching goal of Prof. Lee's research program is to design novel materials with targeted properties using AI-based approaches and to realize them through advanced silicon device fabrication processes.

Highlights

Ph.D at POSTECH, Republic of Korea ('09), Postdoc at Stanford, USA ('10-'13), Assistant Prof. ('13-'17) & Associate Professor ('17-'22) & Professor ('22-'26.8) at Incheon National University, Republic of Korea

144 published papers, 11833 citations, 58 h-index, 43 granted patents, 13 pending patents

Executive Editor of Chemistry of Materials (Sep '25-present) and Associate Editor (Jan '18-Aug '25)

Google Scholar · ORCID

Bio & Research Backgrounds (clickable links below)

Research Achievements (clickable links below)

Activities (clickable links below)

Education

- Ph.D. in Materials Science and Engineering, POSTECH, Republic of Korea
  - Period: 3/2005 - 8/2009
  - Advisor: Professor Hyungjun Kim
  - GPA: 3.7/4.3
  - Thesis: Atomic layer deposition of cobalt
- B.S. in Materials Science and Engineering, Sungkyunkwan University, Republic of Korea
  - Period: 3/1998 - 2/2005 (3/1999 - 5/2002: Military service at the Korean Army)
  - Advisor: Professor Jae Chan Lee
  - GPA: 4.24/4.5, Magna cum laude
  - Thesis: Atomic layer deposition and applications to area-selective growth

Go to the 1st page

Academic & Professional Appointments

- Professor, Graduate School of Semiconductor Materials & Devices Engineering, Ulsan National Institute of Science and Technology (UNIST), Republic of Korea
  - Period: Starting 9/2026
- Professor, Department of Materials Science & Engineering, Incheon National University, Republic of Korea
  - Period: 2/2022 - 8/2026
- Executive Editor, Chemistry of Materials, American Chemical Society Publications, USA
  - Period: 9/2025 - present
- Associate Editor, Chemistry of Materials, American Chemical Society Publications, USA
  - Period: 1/2018 - 8/2025
- Consulting Professor, SK hynix, Republic of Korea
  - Period: Starting 3/2026
- Visiting Professor, Department of Chemical Engineering, Stanford University, USA
  - Period: 2/2024 - 3/2026
- Consulting Professor, SK hynix, Republic of Korea
  - Period: 3/2021 - 2/2022
- Visiting Associate Professor, Department of Chemical Engineering, Stanford University, USA
  - Period: 12/2019 - 2/2021

- Associate Professor, Department of Materials Science & Engineering, Incheon National University, Republic of Korea
  - Period: 3/2017 - 2/2022
- Assistant Professor, Department of Materials Science & Engineering, Incheon National University, Republic of Korea
  - Period: 2/2013 - 2/2017
- Postdoctoral Scholar, Department of Chemical Engineering, Stanford University, USA
  - Period: 3/2010 - 1/2013
- Postdoctoral Scholar, Department of Electrical and Electronic Engineering, Yonsei University, Republic of Korea
  - Period: 9/2009 - 2/2010

Research Experiences

- Postdoctoral Scholar, Department of Chemical Engineering, Stanford University
  - Period: 3/2010 - 1/2013
  - Fund: Department of Energy
  - PI: Professor Stacey F. Bent
  - Research area: Ultra thin atomic layer deposition Pt film for fuel cell (w/ National Renewable Energy Laboratory)
- Postdoctoral Scholar, School of Electrical and Electronic Engineering, Yonsei University
  - Period: 9/2009 - 2/2010
  - Fund: BK21 program
  - PI: Professor Hyungjun Kim
  - Research area: Atomic layer deposition of Co and Ni for metal silicides
- Research Assistant, Dept. of MSE, POSTECH
  - Period: 3/2005 - 8/2009
  - Academic advisor: Professor Hyungjun Kim
  - Research area: Atomic layer deposition of Ru for gate electrode; Atomic layer deposition of Co and Ni for metal silicide contacts; Supercritical fluid deposition of Ru and SiO_2
- Visiting Researcher, Department of Biomedical Engineering, UC Irvine.
  - Period: 2/2007 - 2/2007
  - PI: Professor Noo Li Jeon (Jeon's Microfluidic Lab.)
  - Research area: Bio applications of Cobalt nanostructures
- Visiting Researcher, SNTEK Co. Ltd., Kyunggido, Republic of Korea.
  - Period: 1/2005 - 2/2005
  - PI: Kyung Joon Ahn, CEO
  - Research area: Development and fabrication of atomic layer deposition-magnetron sputtering cluster systems
- Undergraduate Researcher, Dept. of MSE, Sungkyunkwan University
  - Period: 9/2004 - 11/2004
  - PI: Professor Jae Chan Lee (Semiconductor Thin Film Devices Lab.)
  - Research area: Atomic layer deposition and application to area-selective growth.

Go to the 1st page

Technical Skills

- Thin film deposition equipment
  - Extensive experience with the design, construction, and film growth in atomic layer deposition
  - Nanoscale film depositions by sputtering and evaporation
  - Surface treatment and modification by using plasma or coating monolayer
  - Handling of toxic gases and safety system (design and construction of scrubber and gas lines)
- Thin film analysis
  - TEM sample preparation and imaging technique with energy dispersive spectroscopy and electron energy loss spectroscopy analyses
  - X-ray diffraction, synchrotron radiation XRD and XRR with simulation
  - Scanning electron microscopy with energy dispersive spectroscopy
  - Design and construction of in situ surface analysis equipments using quadrupole mass spectroscopy and synchrotron radiation X-ray reflectivity
  - Atomic force microscopy and roughness analyses
  - Secondary ion mass spectroscopy for depth profile analysis
  - Rutherford backscattering and simulation for depth profile analysis
  - X-ray photoelectron spectroscopy for chemical qualification and quantification analyses
  - Auger electron spectroscopy for depth profile and area mapping analyses
  - Glow discharge optical emission spectroscopy for depth profile analysis
  - Surface contact angle measurement for hydrophobicity analysis
  - Inductively-coupled plasma mass spectroscopy for metal quantification
- Device fabrication
  - Rapid thermal annealing (forming gas annealing, oxidation, reduction process)
  - Photolithography
  - Wet chemical etching and reactive ion etching
- Electrical property measurement:
  - Capacitance-voltage (CV) and current-voltage (IV) measurements for metal-oxide-semiconductor characterizations
  - 4-point probe measurement for thin film resistivity
- Software tools:
  - Mathematica for calculation and fitting data
  - Nanoparticle analysis using computational method (ImageJ, Gatan Microscope)

- LabVIEW for control of equipment and data acquisition (Certificates of "LabVIEW Basics II Course" and "Signal Conditioning Course" from National Instrument)
- SPEC and C-PLOT for control of X-ray goniometer and data acquisition of X-ray diffraction
- Parratt32 for X-ray reflectivity simulation, RUMP for Rutherford backscattering simulation

Go to the 1st page

Research Group

CURRENT MEMBERS

Postdoctoral Scholars

- Dr. Aravind H. Patil, Postdoctoral Scholar (Korea Polar Research Institute (KOPRI) & Dept. of Materials Science Engineering, INU, January 2022 – August 2026 → KOPRI & Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Superhydrophobic/Icephobic coatings, Atomic layer deposition
- Dr. Summal Zoha, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, June 2026 – present)
  - Area selective ALD
- Dr. Ngoc Le Trinh, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, June 2026 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Atomic layer modulation, Density functional theory

Ph.D. Students

- Zunair Masroor, Ph.D. Course (Dept. of Materials Science Engineering, INU, September 2022 – present)
  - Molecular dynamics, Deep learning, Density functional theory, AI-driven ALD
- Kyeongmin Min, Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Hot wire ALD, Area selective ALD
- Byungchan Lee, Ph.D. Course (Dept. of Materials Science Engineering, INU, June 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Area selective ALD, AI-driven ALD, Deep learning
- Hamna Fatima, Ph.D. Course (Dept. of Materials Science Engineering, INU, September 2024 – present)
  - Area selective ALE

- Wonjoong Kim, Ph.D. Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2026 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Atomic layer modulation, Kinetic MC simulation, Density functional theory, AI-driven ALD

MS Students

- Yoonseo Choi, MS Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - TiN ALD reactant chemistry, Atomic layer modulation of metal alloys
- Kwangyong An, MS Course (Dept. of Materials Science Engineering, INU, August 2025 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Mo ALD, InN ALD
- Minhyeok Lee, MS Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2025 – present)
  - Kinetic MC simulation, AI-driven ALD
- Dohyun Kim, MS Course (Dept. of Intelligent Semiconductor Engineering, INU, March 2026 – August 2026 → Graduate School of Semiconductor Materials & Devices Engineering, UNIST, September 2026 – present)
  - Kinetic MC simulation, AI-driven ALD

Undergraduate Internship

- Nayoung Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2025 – present)
  - High crystallinity in ultrathin films

ALUMNI

Ph.D. Students

- Dr. Bonwook Gu (now at SK hynix), Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2022 – July 2026), Metal ALD, kinetic MC simulation, AI-driven ALD
- Dr. Chi Thang Nguyen (now at IFW Dresden), Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2018 – August 2023), Area selective ALD, TiO_2
- Dr. Sumaira Yasmeen (now at University of Colorado at Boulder), Ph.D. Course (Dept. of Materials Science Engineering, INU, September 2018 – September 2022), Selective deposition of Ru and SiO_2

- Dr. Hyun Gu Kim (now at Wonik IPS), Ph.D. Course (Dept. of Materials Science Engineering, INU, March 2014 – August 2019), Area selective ALD, ALD of Pt on reduced graphene oxide
- Dr. Jong Seo Park, Ph.D. Course (Co-advising with Prof. Hyungjun Kim at Yonsei University, September 2015 – August 2019), HfO_2 ALD for ferroelectricity

MS Students

- Hye Won Park, MS Course (Dept. of Materials Science Engineering, INU, March 2024 – December 2025), TiN ALD
- Dabin Gong (now at ASM Korea), BS/MS Co-terminal Course (Dept. of Materials Science Engineering, INU, September 2022 – February 2025), ALD, AS-ALD
- Min Kyu Lee (now at Samsung Electronics), MS Course (Dept. of Materials Science Engineering, INU, March 2022 – August 2024), ALD, ASD
- Chan Hui Mun (now at Samsung Electronics), BS/MS Co-terminal Course (Dept. of Materials Science Engineering, INU, March 2020 – February 2022), Superhydrophobic coating, icephobic coating
- Byung Kook Ko (now at Samsung Electronics), MS Course (Dept. of Materials Science Engineering, INU, March 2019 – February 2021), Hf-based ferroelectric materials, atomic layer modulation

Postdoctoral Scholars

- Dr. Abu Saad Ansari, Research Professor (Dept. of Materials Science Engineering, INU, March 2021 – October 2021), DFT, Area selective ALD
- Dr. Rizwan Khan, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, March 2017 – May 2021), Area selective ALD
- Dr. Houda Gaiji, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, July 2019 – May 2020), Superhydrophobic coating for icephobic property
- Dr. Jae Hong Yun, Postdoctoral Scholar (Dept. of Materials Science Engineering, INU, March 2019 – October 2019), Ru ALD, superhydrophobic coating

Undergraduate Internship

- You Chan Ko, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2025 – May 2026), Hot wire ALD, ASD
- Min Sung Kim (now at Applied Materials Korea), Undergraduate Internship (Dept. of Materials Science Engineering, INU, February 2020 – November 2022), MC Simulation for precursor adsorption

- Hyuna Bang, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2021 – March 2022), ALD, AS-ALD
- Soohyun Choi, Undergraduate Internship (Dept. of Materials Science Engineering, INU, February 2021 – August 2021), Area selective ALD
- Min Hee Shong, Undergraduate Internship (Dept. of Materials Science Engineering, INU, March 2021 – June 2021), Hf-based ferroelectric materials, atomic layer modulation
- A Young Lim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, February 2020 – December 2020), MC Simulation for precursor adsorption
- Joonhyuk Ko, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – February 2021), Superhydrophobic coating
- Hyeonseok Oh, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – February 2021), Atomic layer modulation, kinetic MC simulation
- Geonwoo Park, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – February 2021), MC Simulation for precursor adsorption
- Seokjun Seo, Undergraduate Internship (Dept. of Materials Science Engineering, INU, July 2020 – January 2021), Superhydrophobic coating
- Jinsic Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, March 2021 – January 2021), Area selective ALD
- Sang Hui Han, Undergraduate Internship (Dept. of Materials Science Engineering, INU, November 2019 – June 2020), Area selective ALD
- Jongwoo Shin, Undergraduate Internship (Dept. of Materials Science Engineering, INU, March 2019 – February 2020), Selective deposition by ALD
- Je Young Ha, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2019 – February 2020), Area selective ALD
- Se Hyun Shong, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2019 – February 2020), Area selective ALD
- Ho Yeon Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2018 – August 2019), Area selective ALD
- Kyung Taek Park, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2018 – August 2019), Area selective ALD
- Woo Hyuk Kwon, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2016 – December 2018), Electronic textiles, area selective ALD
- Jae Kwang Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2017 – December 2018), Superhydrophobic coating, area selective ALD

- Tae Hwan Im, Undergraduate Internship (Dept. of Materials Science Engineering, INU, October 2017 – November 2018), Area selective ALD
- Hye Jin Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2017 – November 2018), Superhydrophobic coating
- Jae Hee Jung, Undergraduate Internship (Dept. of Materials Science Engineering, INU, January 2018 – August 2018), Area selective atomic layer deposition
- Seung Jae Ahn, Undergraduate Internship (Dept. of Materials Science Engineering, INU, January 2018 – November 2018), Area selective atomic layer deposition
- Yong Woon Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, June 2017 – April 2018), Ferroelectric tunnel junction by ALD
- Yung Hak Lee, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2016 – February 2018), Area selective ALD
- Tae Hee Han, Undergraduate Internship (Dept. of Materials Science Engineering, INU, December 2016 – August 2017), Electronic textiles, smart textile heater
- Sun Hwa Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2015 – December 2016), Chemical bath deposition of rare earth oxides for hydrophobic coating
- Tae Hoon Park, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2014 – December 2015), ALD of Pt on carbon powder, ALD of Pt on reduced graphene oxide
- Min Ji Kim, Undergraduate Internship (Dept. of Materials Science Engineering, INU, September 2014 – December 2015), ALD of TiO_2 for photocatalyst

Go to the 1st page

Journal Articles

SCI & SCIE (144 PUBLISHED, 4 SUBMITTED, 11833 CITATIONS, 58 H-INDEX)

1. "Deep-Learning-Accelerated Dopant Selection for High-k HfO_2 Dielectrics: A Disorder-Resolved Study of Y, Si and Al," Zunair Masroor, Bonwook Gu, Wonjoong Kim, Ngoc Le Trinh, Summal Zoha, and Han-Bo-Ram Lee*, submitted to Journal of Chemical Theory and Computation, [Corresponding author]
2. "Orientation-controlled Area-selective Deposition of Near-single-grain Ruthenium Interconnects," Eun-Hyoung Cho*, Dongmin Kim, Iaan Cho, Bonggeun Shong, Kyeongmin Min, and Han-Bo-Ram Lee, submitted to Nature Nanotechnology, [Co-author]
3. "Transparent and Photovoltaic Water–Semiconductor Interface for Ultra-Low-Light Omnidirectional Photodetection," Sanh Vo Thi, Malkeshkumar Patel, Sourov Hossain, Md Arifur Rahman Barno, Ngoc Le Trinh, Han-Bo-Ram Lee, Yu Kwon Kim*, and Joondong Kim*, Energy & Environmental Materials, e70500, online published, 8/18/2026, https://doi.org/10.1002/eem2.70500, [Co-author]
4. "Synergistic Reinforcement between Al and Si," Han-Bo-Ram Lee*, Chemistry of Materials, 38 (14), 7043-7046 (2026), 7/28/2026, https://doi.org/10.1021/acs.chemmater.6c01759, [Corresponding author]
5. "Chemically Distinct Adsorption of Sulfide Inhibitors for Multi-Surface Passivation and Area Selective Deposition," Summal Zoha†, Fabian Pieck†, Bonwook Gu, Zunair Masroor, Ralf Tonner-Zech*, and Han-Bo-Ram Lee*, Small, e74721, online published, 7/19/2026, https://doi.org/10.1002/smll.74721, †Equal contribution, [Corresponding author]
6. "Atomic Layer Deposition of Titanium Nitride: Reactant-Dependent Ligand Removal and Oxygen Incorporation," Yoonseo Choi, Byungchan Lee, Ngoc Le Trinh, Hyewon Park, and Han-Bo-Ram Lee*, submitted to Journal of Physical Chemistry C, [Corresponding author]
7. "Mechanistic Atomic Hydrogen Chemistry for Ruthenium Deposition: From Ligand Elimination to Area-Selective Patterning," Kyeongmin Min, Chi Thang Nguyen, Eun-Hyoung Cho, Minhyeok Lee, Wonjoong Kim, Youngmin Sunwoo, Eunhyeok Heo, Byungjo Kim, Hao Van Bui and Han-Bo-Ram Lee*, Journal of the American Chemical Society, 148 (27), 28484-28497 (2026), 6/9/2026, https://doi.org/10.1021/jacs.6c04779, [Corresponding author]
8. "Overcoming the Critical Thickness Limit: Interfacial Control of Crystallization Pathways in Atomic-Scale Dielectric Thin Films," Ngoc Le Trinh, Wonjoong Kim, Minhyeok Lee, Bonwook Gu, Sanh Vo Thi, Ji Liu, Michael Nolan, Hyun-Mi Kim, Hyeongkeun Kim, Joondong Kim, Youngho

Kang, and Han-Bo-Ram Lee*, ACS Nano, 20 (24), 17273-17285 (2026), 6/8/2026, https://doi.org/10.1021/acsnano.6c00649, [Corresponding author]

9. "Defect Engineering and Improving Dielectric Performance of Y-Doped ZrO_2 Ultrathin Film through Atomic Layer Modulation," Ngoc Le Trinh, Wonjoong Kim, Minhyeok Lee, Nayoung Lee, Zunair Masroor, Byungha-Kwak, Hyun-Mi Kim, Hyeongkeun Kim, Il-Kwon Oh, and Han-Bo-Ram Lee, submitted to Applied Surface Science, [Corresponding author]
10. "A Dual-Si Atomic Precursor Enabling High-Quality SiN_x Thin Films via Very High-Frequency Plasma Atomic Layer Deposition," Young-Jin Lim, Yujin Lee, Kim-Hue Thi Dinh, Min-Jeong Rhee, Yea-Ji Kim, Ngoc Le Trinh, Bonwook Gu, Youngho Kang, Jae Hack Jeong, Han-Bo-Ram Lee*, Stacey Bent*, and Il-Kwon Oh*, Chemistry of Materials, 38 (11), 5679–5690 (2026), 5/16/2026, https://doi.org/10.1021/acs.chemmater.6c00586, [Corresponding author]
11. "AI-Driven Inverse Design of Complex Oxide Thin Films for Semiconductor Devices: A Case Study of Hf–Zr–O," Bonwook Gu, Ngoc Le Trinh, Wonjoong Kim, Zunair Masroor, and Han-Bo-Ram Lee*, Chemistry of Materials, 38 (10), 5285–5297 (2026), 5/11/2026, https://doi.org/10.1021/acs.chemmater.6c00978, [Corresponding author]
12. "Atomic Layer Deposition of Molybdenum Carbide and Substrate-dependent Reduction to Metallic Molybdenum," Kwangyong An, Kieran G. Lawford, Bonwook Gu, Aravind H. Patil, Aaron D. Rogers, Seán T. Barry, and Han-Bo-Ram Lee*, Journal of Materials Chemistry C, 14 (23), 10006-10019 (2026), 4/14/2026, https://doi.org/10.1039/D6TC00454G, [Corresponding author]
13. "Solar-Driven Icephobic Layer Design by Superhydrophobic Materials," Aravind H. Patil, Sumin An, Youngho Kang, Sawanta S. Mali, Chang Kook Hong, Joohan Lee, Changhyun Chung*, and Han-Bo-Ram-Lee*, Applied Surface Science, 732, 166580 (2026), 3/13/2026, https://doi.org/10.1016/j.apsusc.2026.166580, [Corresponding author]
14. "Tuning Surface Reactivity Pathways through Molecular Inhibitor Redosing for Precision Nanopatterning," Byungchan Lee, Chi Thang Nguyen, Minhyeok Lee, Ngoc Le Trinh, Kyeongmin Min, Younho Kang, Eun-Hyoung Cho, and Han-Bo-Ram Lee, Materials Horizon, 13 (10), 5035-5049 (2026), 3/3/2026, https://doi.org/10.1039/D5MH02445E, [Corresponding author]
15. "Cobalt Precursor as Surface Inhibitor for Area-Selective Atomic Layer Deposition of Nanoscale Electronic Interconnects," Zunair Masroor, Ngoc Le Trinh, Summal Zoha, Youngho Kang, and Han-Bo-Ram Lee*, ACS Applied Nano Materials, 9 (4), 1869-1878 (2026), 1/21/2026, https://doi.org/10.1021/acsanm.5c05042, [Corresponding author]
16. "Atomic Layer Modulation of Ruthenium Aluminum Oxide through Reactivity Control of Precursors for Seedless Copper Interconnects," Chi Thang Nguyen, Miso Kim, Youn-Hye Kim, Taehoon Cheon, Ngoc Le Trinh, Sehee Kim, Soo-Hyun Kim, Bonggeun Shong*, and Han-Bo-

Ram Lee*, Chemistry of Materials, 37 (24), 9745-9757 (2025), 11/27/2025, https://dx.doi.org/10.1021/acs.chemmater.5c02156, [Corresponding author]

17. "Area Selective Deposition of Tungsten and Ruthenium using a Tungsten Precursor Inhibitor," Bonwook Gu, Chi Thang Nguyen, Mingyu Lee, Sunghwan Jo, Ngoc Le Trinh, Youngho Kang, Seung Wook Ryu and Han-Bo-Ram Lee*, Applied Surface Science, 720, 165129 (2026), 11/6/2025, https://doi.org/10.1016/j.apsusc.2025.165129, [Corresponding author]
18. "Materials Chemistry in the Far South," Han-Bo-Ram Lee*, 37 (20), 8103-8106 (2025), Chemistry of Materials, http://doi.org/10.1021/acs.chemmater.5c02391, [Corresponding author]
19. "Functionalization of Si_3N_4 with Aldehyde Inhibitor for Area-Selective Deposition of HfO_2 and SiO_2," Byung Chan Lee, Seoung Won Shim, Ngoc Le Trinh, Aravind H. Patil, Chi Thang Nguyen, Taehoon Cheon, Soo-Hyun Kim, Youngho Kang, and Han-Bo-Ram Lee*, Journal of Materials Chemistry C, 13 (37), 19316-19329 (2025), 8/11/2025, https://doi.org/10.1039/D5TC01969A, [Corresponding author]
20. "Ultrathin Ag Film of Transparent Electrode for Flexible Schottky Photodetection and Enhanced Charge Transport," Sanh Vo Thi, Malkeshkumar Patel, Thanh Tai Nguyen, Sourov Hossain, Seoyoung Lim, Ngoc Le Trinh, Han-Bo-Ram Lee, Dong-Wook Kim*, and Joondong Kim*, ACS Applied Electronic Materials, 7(9), 4339-4351 (2025), 4/8/2025, https://doi.org/10.1021/acsaelm.5c00498, [Co-author]
21. "Synergistic Effects of Ionic Liquid Integrated Superhydrophobic Composite Coatings on Anti-Icing Properties," Aravind H. Patil, Ngoc Le Trinh, Hackwon Do, Youngho Kang, Joohan Lee, Changhyun Chung*, and Han-Bo-Ram Lee*, 693, 162749 (2025), 2/21/2025, Applied Surface Science, https://doi.org/10.1016/j.apsusc.2025.162749, [Corresponding author]
22. "Atomic-level Stoichiometry Control of Ferroelectric $Hf_xZr_yO_z$ Thin Films by Understanding Molecular-level Chemical Physical Reactions," Ngoc Le Trinh, Bonwook Gu, Kun Yang, Chi Thang Nguyen, Byungchan Lee, Hyun-Mi Kim, Hyeongkeun Kim, Youngho Kang, Min Hyuk Park*, and Han-Bo-Ram Lee*, ACS Nano, 19(3), 3562-3578 (2025), 1/15/2025, https://doi.org/10.1021/acsnano.4c13595, [Corresponding author]
23. "Insights into the Thermal Decomposition Mechanism of Molybdenum(VI) Iminopyridine Adducts," Lara K. Watanabe, Aaron D. Rogers, Marshall T. Atherton, Kieran G. Lawford, Bonwook Gu, Supill Chun, Sanghyun Lee, Dongkyun Lee, Han-Bo-Ram Lee, and Seán T. Barry, Chemistry of Materials, 36(24), 11985-11995 (2024), 12/5/2024, https://doi.org/10.1021/acs.chemmater.4c02689, [Co-author]
24. "Area-Selective Atomic Layer Deposition of Ruthenium via Reduction of Interfacial Oxidation," Eun-Hyoung Cho, Dabin Kong, Iaan Cho, Youngchul Leem, Young Min Lee, Miso Kim, Chi Thang

Nguyen, Jeong Yub Lee, Bonggeun Shong*, and Han-Bo-Ram Lee*, 36, 18, 8663–8672 (2024), 9/13/2024, Chemistry of Materials, https://doi.org/10.1021/acs.chemmater.4c01133, [Corresponding author]

25. "Si precursor Inhibitors for Area Selective Deposition of Ru," Bonwook Gu, Sumaira Yasmeen, Geun-Ha Oh, Il-Kwon Oh, Youngho Kanga, and Han-Bo-Ram Lee*, Applied Surface Science, 669, 1650530 (2024), 6/12/2024, https://doi.org/10.1016/j.apsusc.2024.160530, [Corresponding author]
26. "Organosulfide Inhibitor Instigated Passivation of Multiple Substrates for Area-Selective Atomic Layer Deposition of HfO_2," Summal Zoha, Fabian Pieck, Bonwook Gu, Ralf Tonner-Zech*, and Han-Bo-Ram Lee*, Chemistry of Materials, 36 (6), 2661-2673 (2024), 2/22/2024, https://dx.doi.org/10.1021/acs.chemmater.3c02525, [Corresponding author]
27. "Facile Approach for Designing Icephobic Coatings using Polymers/Silica Nanoparticle Composites via Self-Formation of Superhydrophobic Surfaces," Aravind H. Patil, Ngoc Le Trinh, Hackwon Do, Houda Gaiji, Youngho Kang, Joohan Lee, Changhyun Chung*, and Han-Bo-Ram Lee*, Advanced Materials Interfaces, 2300919, 1-13 (2024), 2/7/2024, https://doi.org/10.1002/admi.202300919, [Corresponding author]
28. "Area Selective Deposition of Ruthenium Using Homometallic Precursor Inhibitor," Chi Thang Nguyen, Eun-Hyoung Cho, Ngoc Le Trinh, Bonwook Gu, Mingyu Lee, Sunghee Lee, Jeong Yub Lee, Youngho Kang, and Han-Bo-Ram-Lee*, Chemistry of Materials, 35(14), 5331-5340 (2023), 7/3/2023, https://doi.org/10.1021/acs.chemmater.3c00525, [Corresponding author]
29. "Borderless Collaboration in Materials Chemistry for an Industrial Ecosystem in the Semiconductor Industry," Han-Bo-Ram Lee* and Seiyon Kim, Chemistry of Materials, 35(11), 4117-4119 (2023), 6/13/2023, https://doi.org/10.1021/acs.chemmater.3c01077, [Corresponding author]
30. "Expanding Atomic Layer Deposition from Silicon Substrates to Coke Bottles," Han-Bo-Ram Lee,* Carlos Toro,* and Sara E. Skrabalak*, Chemistry of Materials, 35(9), 3343-3344 (2023), 5/9/2023, https://doi.org/10.1021/acs.chemmater.3c00900, [Corresponding author]
31. "Self-formation of a Ru/ZnO Multifunctional Bilayer through Area-Selective Atomic Layer Deposition for Interconnect Technology," Yuki Mori, Taehoon Cheon, Yohei Kotsugi, Youn-Hye Kim, Yejin Park, Mohd Zahid Ansari, Debananda Mohapatra, Yujin Jang, Jong-Seong Bae, Woobin Kwon, Gahui Kim, Young-Bae Park, Han-Bo-Ram Lee, Wooseok Song,* and Soo-Hyun Kim*, Small, 2300290 (2023), 5/1/2023, https://doi.org/10.1002/smll.202300290, [Co-author]
32. "Dysprosium Incorporation for Phase Stabilization of Atomic-Layer-Deposited HfO_2 Thin Films," Yujin Lee, Kangsik Kim, Zonghoon Lee, Hong-Sub Lee, Han-Bo-Ram Lee, Woo-Hee Kim*, Il-

Kwon Oh*, and Hyungjun Kim*, Chemistry of Materials, 35(6), 2312-2320 (2023), 3/16/2023, https://doi.org/10.1021/acs.chemmater.2c02862, [Co-author]

33. "Gradient Area-Selective Deposition for Seamless Gap-filling in 3D Nanostructures through Surface Chemical Reactivity Control," Chi Thang Nguyen, Eun-Hyoung Cho, Bonwook Gu, Sunghee Lee, Hae-Sung Kim, Jeongwoo Park, Neung-Kyung Yu, Sangwoo Shin, Bonggeun Shong, Jeong Yub Lee, and Han-Bo-Ram-Lee*, Nature Communications, 13, 9579 (2022), 12/9/2022, https://doi.org/10.1038/s41467-022-35428-6, [Corresponding author]

34. "Computational Modeling of Physical Surface Reactions of Precursors in Atomic Layer Deposition by Monte Carlo Simulations on Home Desktop Computer," Bonwook Gu, Ngoc Le Trinh, Chi Thang Nguyen, Sumaira Yasmeen, Youngho Kang, and Han-Bo-Ram-Lee*, Chemistry of Materials, 34(17), 7635–7649 (2022), 8/26/2022, https://doi.org/10.1021/acs.chemmater.2c00854, [Corresponding author]

35. "Atomic Layer Deposition Beyond Thin Film Deposition Technology," Sumaira Yasmeen, Seung Wook Ryu, Sung-Hoon Lee, and Han-Bo-Ram-Lee*, Advanced Materials Technologies, 2200876 (2022), 10/10/2022, https://doi.org/10.1002/admt.202200876, [Corresponding author]

36. "Elucidating Reaction Mechanism of Atomic Layer Deposition of Al_2O_3 with a Series of $Al(CH)_xCl_{3-x}$ and $Al(C_yH_{2y+1})_3$ Precursors," Il-Kwon Oh, Tania E. Sandoval, Tzu-Ling Liu, Nathan Richey, Chi Thang Nguyen, Bonwook Gu, Han-Bo-Ram Lee, Ralf Tonner-Zech, and Stacey F. Bent*, Journal of the American Chemical Society, 144(6), 11757-11766 (2022), 6/6/2022, https://doi.org/10.1021/jacs.2c03752, [Co-author]

37. "Icephobic Coating through Self-formed Superhydrophobic Surface using Polymer and Microsize Particles," Chan Hui Moon, Sumaira Yasmeen, Kiho Park, Houda Gaiji, Changhyun Chung, Hyoungkwon Kim, Hyoung-Seok Moon, Jang Wook Choi, and Han-Bo-Ram-Lee*, ACS Applied Materials and Interfaces, 14(2), 3334–3343 (2022), 1/4/2022, https://doi.org/10.1021/acsami.1c22404, [Corresponding author]

38. "Growth modulation of atomic layer deposition of HfO_2 by combinations of H_2O and O_3 reactants," Byeong Guk Ko, Chi Thang Nguyen, Bonwook Gu, Mohammad Rizwan Khan, Kunwoo Park, Hongjun Oh, Jungwon Park, Bonggeun Shong, Han-Bo-Ram Lee*, Dalton Transactions, 50 (48), 17935-17944 (2021), 11/23/2021, https://doi.org/10.1039/d1dt03465k, [Corresponding author]

39. "Evaluation of Silicon Tetrahalide Precursors for Low-temperature Thermal Atomic Layer Deposition of Silicon Nitride," Neung-Kyung Yu, Chan Hui Moon, Jeongwoo Park, Han-Bo-Ram Lee*, and Bonggeun Shong*, Applied Surface Science, 565, 150603, 7/12/2021, https://doi.org/10.1016/j.apsusc.2021.150603, [Corresponding author]

40. "Toward Enhanced Humidity Stability of Triboelectric Mechanical Sensors via Atomic Layer Deposition," Wook Kim, Sumaira Yasmeen, Chi Thang Nguyen, Han-Bo-Ram Lee*, Dukhyun Choi*, Nanomaterials, 11(7), 1795, 7/9/2021, https://doi.org/10.3390/nano11071795, [Corresponding author]

41. "Atomic Layer Modulation of Multicomponent Thin Films through Combination of Experimental and Theoretical Approaches," Chi Thang Nguyen, Bonwook Gu, Taehoon Cheon, Jeongwoo Park, Mohammad Rizwan Khan, Soo-Hyun Kim, Bonggeun Shong, and Han-Bo-Ram Lee*, Chemistry of Materials, 33(12), 4435–4444 (2021), 06/10/2021, https://doi.org/10.1021/acs.chemmater.1c00508, [Corresponding author]

42. "Self-Formation of Superhydrophobic Surfaces through Interfacial Energy Engineering between Liquids and Particles," Sumaira Yasmeen†, Jaehong Yoon†, Chan Hui Moon, Mohammad Rizwan Khan, Houda Gaiji, Sangwoo Shin, Il-Kwon Oh*, and Han-Bo-Ram Lee*, Langmuir, 37(17), 5356-5363 (2021), 4/23/2021, https://doi.org/10.1021/acs.langmuir.1c00481, [Corresponding author]

43. "Promoting Lithium Electrodeposition Towards the Bottom of 3-D Copper Meshes in Lithium-based Batteries," Sujung Kim, Yena Kim, Taegyu Jang, Chi Thang Nguyen, Han-Bo-Ram Lee, and Hye Ryung Byon, Journal of Power Source, 472, 228495, 10/1/2020, https://doi.org/10.1016/j.jpowsour.2020.228495 [Co-author]

44. "Effects of Al Precursors on Deposition Selectivity of Atomic Layer Deposition of Al_2O_3 using Ethanethiol Inhibitor," Hyun Gu Kim, Miso Kim, Bonwook Gu, Mohammad Rizwan Khan, Byeong Guk Ko, Sumaira Yasmeen, Chang Su Kim, Se-Hun Kwon, Jiyong Kim, Junhyuck Kwon, Kwangseon Jin, Byungchul Cho, J. -S. Chun, Bonggeun Shong*, and Han-Bo-Ram Lee*, Chemistry of Materials, 32 (20), 8921-8929 (2020), 9/28/2020, https://doi.org/10.1021/acs.chemmater.0c02798 [Corresponding author]

45. "Atomic Level Deposition to Extend Moore's Law and Beyond," Rong Chen*, Yi-Cheng Li, Jia-Ming Cai, Kun Cao, and Han-Bo-Ram Lee, International Journal of Extreme Manufacturing, 2 (2), 022002-1 (2020), 4/30/2020, https://doi.org/10.1088/2631-7990/ab83e0, [Co-author]

46. "A Selective Toolbox for Nanofabrication," Han-Bo-Ram Lee* and Stacey F. Bent, Chemistry of Materials, 32 (8), 3323-3324 (2020), 4/14/2020, https://dx.doi.org/10.1021/acs.chemmater.0c00838, [Corresponding author]

47. "Ultrathin Effective TiN Protective Films Prepared by Plasma-enhanced Atomic Layer Deposition for High Performance Metallic Bipolar Plates of Polymer Electrolyte Membrane Fuel Cells," Woo-Jae Lee, Eun-Young Yun, Han-Bo-Ram Lee, Suck Won Hong*, and Se-Hun Kwon*, Applied

Surface Science, 519, 146215 (2020), 3/27/2020, https://doi.org/10.1016/j.apsusc.2020.146215, [Co-author]

48. "Tunable Color Coating of E-Textiles by the Atomic Layer Deposition of Multilayer TiO_2/Al_2O_3 Films," Mohammad Rizwan Khan, Hyun Gu Kim, Jong Seo Park, Jong Woo Shin, Chi Thang Nguyen, and Han-Bo-Ram Lee*, Langmuir, 36 (11), 2794-2801 (2020), 2/13/2020, https://dx.doi.org/10.1021/acs.langmuir.9b03988, [Corresponding author]
49. "Preparation of a Hydrophobic Cerium Oxide Nanoparticle Coating with Polymer Binder via a Facile Solution Route," Sumaira Yasmeen, Mohammad Rizwan Khan, Kiho Park, Yunshik Cho, Jang Wook Choi, Hyoung-Seok Moon, and Han-Bo-Ram Lee*, Ceramic International, 46 (8), 12209-12215 (2020), 2/3/2020, https://doi.org/10.1016/j.ceramint.2020.01.268, [Corresponding author]
50. "Surface Energy Change of Atomic-scale Metal Oxide Thin Films by Phase Transformation," Il-Kwon Oh, Li Zeng, Jae-Eun Kim, Jong-Seo Park, Kangsik Kim, Hyunsoo Lee, Seunggi Seo, Mohamed Rizwan Khan, Sangmo Kim, Chung Wung Park, Junghoon Lee, Bonggeun Shong, Zonghoon Lee, Stacey F. Bent, Hyungjun Kim, Jeong Young Park, and Han-Bo-Ram Lee*, ACS Nano, 14(1), 676-687 (2020), 1/13/2020, https://doi.org/10.1021/acsnano.9b07430, [Corresponding author]
51. "Effect of Molecular Backbone Structure on Gas Phase Coupling Chemistry of (Thio)urea/urethane," Shimeles Shumi Raya, Abu Saad Ansari, Sung Gu Kang, Han-Bo-Ram Lee* and Bonggeun Shong*, Progress in Organic Coating, 140, 105509 (2020), https://doi.org/10.1016/j.porgcoat.2019.105509, [Corresponding author]
52. "Reaction Mechanism of Pt Atomic Layer Deposition on Various Textile Surfaces," Il-Kwon Oh, Jong S. Park, Mohammad Rizwan Khan, Kangsik Kim, Zonghoon Lee, Bonggeun Shong, and Han-Bo-Ram Lee*, Chemistry of Materials, 31 (21), 8995-9002 (2019), 10/23/2019, http://doi.org/10.1021/acs.chemmater.9b03171, [Corresponding author]
53. "Science against Pseudoscience," Han-Bo-Ram Lee*, Chemistry of Materials, 31 (18), 7113-7115 (2019), 9/24/2019, https://doi.org/10.1021/acs.chemmater.9b03134, [Corresponding author]
54. "Moisture Barrier Properties of Low-temperature Atomic Layer Deposited Al_2O_3 using Various Oxidants," Taewook Nam, Haksoo Lee, Seunggi Seo, Sung Min Cho, Bonggeun Shong, Han-Bo-Ram Lee, and Hyungjun Kim*, Ceramic International, 45 (15), 19105-19112 (2019), 6/20/2019, https://doi.org/10.1016/j.ceramint.2019.06.156, [Co-author]
55. "Thermal Atomic Layer Deposition of Metallic Ru using H_2O as a Reactant," Chi Thang Nguyen, Jaehong Yoon, Rizwan Khan, Bonggeun Shong* and Han-Bo-Ram Lee*, Applied Surface

Science, 488, 896-902 (2019), 5/21/2019, https://doi.org/10.1016/j.apsusc.2019.05.242 [Corresponding author]

56. "Analysis of Defect Recovery in Reduced Graphene Oxide and Its Application as a Heater for Self-Healing Polymers," Hyun Gu Kim, Il-Kwon Oh, Seungmin Lee, Sera Jeon, Hyunyong Choi, Kwanpyo Kim, Joo Ho Yang, Jae Woo Chung, Jaekwang Lee, Woo-Hee, Kim*, and Han-Bo-Ram Lee*, ACS Applied Materials and Interfaces, 11 (18), 16804-16814 (2019), 4/9/2019, http://doi.org/10.1021/acsami.8b19955 [Corresponding author]

57. "Formulation of PEDOT:S-Graphene Hybrid and Its Application as Transparent Conducting Electrode Materials," Seongho Jo, Hyun Gu Kim, Young Ji Son, Ari Chae, Han-Bo-Ram Lee, and Insik In*, Materials Today: Proceedings, 10, 448-455 (2019), 4/4/2019, https://doi.org/10.1016/j.matpr.2019.03.009, [Co-author]

58. "The Era of Atomic Crafting," Han-Bo-Ram Lee*, Chemistry of Materials, 31 (5), 1471-1472 (2019), 3/12/2019, http://doi.org/10.1021/acs.chemmater.9b00654, [Corresponding author]

59. "Area-Selective Atomic Layer Deposition Using Si Precursors as Inhibitors," Rizwan Khan, Bonggeun Shong, Byeong Guk Ko, Jae Kwang Lee, Hyunsoo Lee, Jeong Young Park, Il-Kwon Oh, Shimeles Shumi Raya, Hyun Min Hong, Kwun-Bum Chung, Erik J. Luber, Yoon-Seok Kim, Chul-Ho Lee, Woo-Hee Kim, and Han-Bo-Ram Lee*, Chemistry of Materials, 30 (21), 7603-7610 (2018), 10/11/2018, http://doi.org/10.1021/acs.chemmater.8b02774 [Corresponding author]

60. "In Honor of Professor Markku Leskelä," Mikko Ritala, Han-Bo-Ram Lee, Jillian Buriak, and Seán T. Barry*, Chemistry of Materials, 30 (14), 4469–4474 (2018), 7/24/2018, http://doi.org/10.1021/acs.chemmater.8b02742, [Corresponding author]

61. "Comparative Study of the Growth Characteristics and Electrical Properties of Atomic-Layer-Deposited HfO_2 Films obtained from Metal Halide and Amide Precursors," Il-Kwon Oh, Bo-Eun Park, Seunggi Seo, Byung Chul Yeo, Jukka Tanskanen, Han-Bo-Ram Lee, Woo-Hee Kim, and Hyungjun Kim*, Journal of Materials Chemistry C, 6, 7367-7376 (2018), 6/29/2018, https://doi.org/10.1039/C8TC01476K, [Co-author]

62. "Molecular Oxidation of Surface $-CH_3$ during Atomic Layer Deposition of Al_2O_3 with H_2O, H_2O_2, and O_3: A Theoretical Study," Seunggi Seo, Taewook Nam, Han-Bo-Ram Lee, Hyungjun Kim*, and Bonggeun Shong*, Applied Surface Science, 457, 376-380 (2018), 6/19/2018, https://doi.org/10.1016/j.apsusc.2018.06.160, [Co-author]

63. "Effect of h-BN Coating on Nucleate Boiling Heat Transfer Performance in Pool Boiling," Ji Min Kim, ByeongTak Kong, Somchai Wongwises, Han-Bo-Ram Lee, and Ho Seon Ahn*, Experimental Thermal and Fluid Science, 93, 12-19 (2018), 5/14/2018, https://doi.org/10.1016/j.expthermflusci.2018.05.010, [Co-author]

64. “Surface Wettability of Nitrogen-doped TiO_2 Films by Atomic Layer Deposition using NH_4OH as a Doping Source,” Rizwan Khan, Kyung Yong Ko, Jong Seo Park, Hyungjun Kim*, and Han-Bo-Ram Lee*, Nanoscience and Nanotechnology Letters, 10(5-6), 779-783 (2018), 5/1/2018, https://doi.org/10.1166/nnl.2018.2663, [Corresponding author]
65. “Amorphous TiO_2/p-Si Heterojunction Photodiode prepared by Low-temperature Atomic Layer Deposition,” Jaehong Yoon, Rizwan Khan, Il-Kwon Oh, Hyungjun Kim*, and Han-Bo-Ram Lee*, Nanoscience and Nanotechnology Letters, 10(5-6), 779-783 (2018), 5/1/2018, https://doi.org/10.1166/nnl.2018.2638, [Corresponding author]
66. “Circular Double-Patterning Lithography using a Block Copolymer Template and Atomic Layer Deposition,” Zhixin Wan, Ha Jin Lee, Hyun Gu Kim, Gyeong Cheon Jo, Woon Ik Park, Seung Wook Ryu, Han-Bo-Ram Lee*, and Se-Hun Kwon*, Advanced Materials Interfaces, 5 (16), 1800054 (2018), 4/20/2018, https://doi.org/10.1002/admi.201800054, [Corresponding author]
67. “Interlayer-assisted Atomic Layer Deposition of MgO as a Magnetic Tunneling Junction Insulators,” Seung Wook Ryu, Jeong-Gyu Song, Hyun Gu Kim, Hyungjun Kim*, and Han-Bo-Ram Lee*, Journal of Alloys and Compound, 747, 505-510 (2018), 5/30/2018, https://doi.org/10.1016/j.jallcom.2018.03.021, [Corresponding author]
68. “Copper Indium Selenide Water Splitting Photoanodes with Artificially Designed Heterophasic Blended Structure and Their High Photoelectrochemical Performances,” Joo Sung Kim, Seung Ki Baek, Young Been Kim, Hyun Woo Do, Yong Hun Kwon, Sung Woon Cho, Young Dae Yun, Jae Hong Yoon, Han-Bo-Ram Lee, Sang-Woo Kim, and Hyung Koun Cho*, Nano Energy, 46, 1-10 (2018), 1/18/2018, https://doi.org/10.1016/j.nanoen.2018.01.023, [Co-author]
69. “Water-Erasable Memory Device for Security Applications Prepared by the Atomic Layer Deposition of GeO_2,” Chang Mo Yoon, Il-Kwon Oh, Yoo-Jin Lee, Jeong-Gyu Song, Su Jeong Lee, Jae-Min Myoung, Hyun Gu Kim, Hyoung-Seok Moon, Bonggeun Shong, Hyungjun Kim*, and Han-Bo-Ram Lee*, Chemistry of Materials, 30 (3), 830–840 (2018), 1/12/2018, http://doi.org/10.1021/acs.chemmater.7b04371, [Corresponding author]
70. “Effect of Porous Graphene Networks and Micropillar Arrays on Boiling Heat Transfer Performance,” Ji Min Kim, Su Cheong Park, ByeongTak Kong, Han-Bo-Ram Lee, and Ho Seon Ahn*, Experimental Thermal and Fluid Science, 93, 153-164, 2018, 12/28/2017, https://doi.org/10.1016/j.expthermflusci.2017.12.029, [Co-author]
71. “Cobalt Titanium Nitride Amorphous Metal Alloys by Atomic Layer Deposition,” Taewook Nam, Chang Wan Lee, Taehoon Cheon, Woo-Jae Lee, Soo-Hyun Kim, Se-Hun Kwon, Hyungjun Kim*, and Han-Bo-Ram Lee*, Journal of Alloys and Compound, 737, 684-692 (2017), 12/6/2017, http://dx.doi.org/10.1016/j.jallcom.2017.12.023, [Corresponding author]

72. "Reaction Mechanism of Area-Selective Atomic Layer Deposition for Al_2O_3 Nanopatterns," Seunggi Seo† & Il-Kwon Oh†, Byung Chul Yeo, Sang Soo Han, Chang Mo Yoon, Joon Young Yang, Jonggeun Yoon, Choongkeun Yoo, Ho-jin Kim, Yong-baek Lee, Su Jeong Lee, Jae-Min Myoung, Han-Bo-Ram Lee, Woo-Hee Kim, and Hyungjun Kim*, ACS Applied Materials & Interfaces, 9 (47), 41607-41617, 11/7/2017, http://dx.doi.org/10.1021/acsami.7b13365, †Equal contribution, [Co-author]
73. "Conduction Mechanism Change with Transport Oxide Layer Thickness in Oxide Hetero-interface Diode," Bu-il Nam, Jong Seo Park, Keon-Hee Lim, Yong-keon Ahn, Jinwon Lee, Jun-Woo Park, Nam-Kwang Cho, Donggun Lee, Han-Bo-Ram Lee*, and Youn Sang Kim*, Applied Physics Letters, 111, 053506 (2017), 8/2/2017, http://dx.doi.org/10.1063/1.4996862 [Corresponding author]
74. "Dual Role of Sb-incorporated Buffer Layers for High Efficiency Cuprous Oxide Photocathodic Performance: Remarkably Enhanced Crystallinity and Effective Hole Transport," Seung Ki Baek, Joo Sung Kim, Young Bin Kim, Jae Hong Yun, Han-Bo-Ram Lee, and Hyung Koun Cho*, ACS Sustainable Chemistry & Engineering, 5(9), 8213-8221 (2017), 7/17/2017, http://dx.doi.org/10.1021/acssuschemeng.7b01889, [Co-author]
75. "Atomic layer deposition of Y-stabilized ZrO_2 for advanced DRAM capacitors," Bo-Eun Park, Il-Kwon Oh, Chandreswar Mahata, Chang Wan Lee, David Thompson, Han-Bo-Ram Lee, Wan Joo Maeng, and Hyungjun Kim*, Journal of Alloys and Compound, 722, 307-322 (2017), 10/25/2017, https://doi.org/10.1016/j.jallcom.2017.06.036, [Co-author]
76. "Atomic Layer Deposition on 2D Materials," Hyun Gu Kim and Han-Bo-Ram Lee*, Chemistry of Materials, 29(9), 3809-3826 (2017), 4/25/2017, http://dx.doi.org/10.1021/acs.chemmater.6b05103, [Corresponding author]
77. "Nanoconfined Atomic Layer Deposition of TiO_2/Pt Nanocavities: Towards Ultra-Small Highly Efficient Catalytic Nanorockets," Jinxing Li, Wenjuan Liu, Jiyuan Wang, Isaac Rozen, Sha He, Chuanrui Chen, Hyun Gu Kim, Ha-Jin Lee, Han-Bo-Ram Lee, Se-Hun Kwon, Tianlong Li, Longqiu Li*, Yongfeng Mei*, and Joseph Wang*, Advanced Functional Materials, 27, 1700598 (2017), 4/24/2017, http://dx.doi.org/10.1002/adfm.201700598, [Co-author]
78. "Vapor Phase Synthesis of TaS_2 Nanocrystals with Iodine as Transport Agent," Gangtae Jin, Chaeeun Kim, Hyunjin Jo, Se-Hun Kwon, Seong-Jun Jeong, Han-Bo-Ram Lee*, and Ji-Hoon Ahn*, Japanese Journal of Applied Physics, 56, 045501 (2017), 3/3/2017, https://doi.org/10.7567/JJAP.56.045501, [Corresponding author]
79. "Distribution of Oxygen Functional Groups of Graphene Oxide obtained from Low-temperature Atomic Layer Deposition of Titanium Oxide," Dong Seok Shin†, Hyun Gu Kim†, Ho Seon Ahn†,

Hu Young Jeong, Yeon Jung Kim, Dorj Odkhuu, Han-Bo-Ram Lee* and Byung Hoon Kim*, RSC Advances, 7, 13979-13984 (2017), 3/2/2017, http://dx.doi.org/10.1039/C7RA00114B, [Corresponding author]

80. "A Composite Layer of Atomic-Layer-Deposited Al_2O_3 and Graphene for Flexible Moisture Barrier," Taewook Nam, Youngju Park, Seung-Woo Seo, Il-Kwon Oh, Jong-Hyun Ahn, Sung Min Cho, Hyungjun Kim*, and Han-Bo-Ram Lee*, Carbon, 116, 553-561 (2017), 2/17/2017, http://dx.doi.org/10.1016/j.carbon.2017.02.023, [Corresponding author]

81. "Atomic Layer Deposition of 1D and 2D Nickel Nanostructures on Graphite," Seung Wook Ryu† & Jaehong Yoon† & Hyoung-Seok Moon†, Bonggeun Shong, Hyungjun Kim, and Han-Bo-Ram Lee*, Nanotechnology, 28(11), 115301 (2017), 2/13/2017, https://doi.org/10.1088/1361-6528/aa5aec, †Equal contribution, [Corresponding author]

82. "Growth Mechanism of Co Thin Films Formed by Plasma-enhanced Atomic Layer Deposition using NH_3 as Plasma Reactant," Il-Kwon Oh, Hyungjun Kim*, and Han-Bo-Ram Lee*, Current Applied Physics, 17(3), 333-338 (2016), 12/24/2016, http://dx.doi.org/10.1016/j.cap.2016.12.021, [Corresponding author]

83. "Highly Conductive and Flexible Fiber for Textile Electronics Obtained by Extremely Low Temperature Atomic Layer Deposition of Pt," Jaehong Lee† & Jaehong Yoon†, Hyun Gu Kim, Subin Kang, Woo-Suk Oh, Hassan Algadi, Saleh Al-Sayari, Bonggeun Shong, Soo-Hyun Kim, Hyungjun Kim, Taeyoon Lee*, and Han-Bo-Ram Lee*, NPG Asia Materials, 8, e331 (2016), 11/25/2016, http://dx.doi.org/10.1038/am.2016.182, †Equal contribution, [Corresponding author]

84. "Uniform Color Coating of Multilayered TiO_2/Al_2O_3 Films by Atomic Layer Deposition," Woo-Hee Kim, Hyungjun Kim*, and Han-Bo-Ram Lee*, Journal of Coatings Technology and Research, 14(1), 177-183 (2017), 8/29/2016, http://dx.doi.org/10.1007/s11998-016-9840-2, [Corresponding author]

85. "Fabrication of 50-nm Scale Nanostructures by Block Copolymer and Its Characteristics of Surface-enhanced Raman Scattering," Jae Hee Shin, Hyun Gu Kim, Gwang Min Baek, Reehyang Kim, Suwan Jeon, Jeong Ho Mun, Han-Bo-Ram Lee, Yeon Sik Jeong, Sang Ouk Kim, Kyoung Nam Kim, and Geun Young Yeom*, RSC Advance, 6(75), 70756-70762 (2016), 7/26/2016, http://dx.doi.org/10.1039/C6RA08608J, [Co-author]

86. "Very High Frequency Plasma Reactant for Atomic Layer Deposition," Il-Kwon Oh, Gilsang Yoo, Chang Mo Yoon, Tae Hyung Kim, Geun Young Yeom, Kangsik Kim, Zonghoon Lee, Hanearl Jung, Changwan Lee, Hyungjun Kim* and Han-Bo-Ram Lee*, Applied Surface Science, 387, 109-117 (2016), 6/14/2016, http://dx.doi.org/10.1016/j.apsusc.2016.06.048, [Corresponding author]

87. "High Efficiency n-Si/p-Cu_2O Core–shell Nanowires Photodiode Prepared by Atomic Layer Deposition of Cu_2O on Well-ordered Si Nanowires Array," Hangil Kim, Soo-Hyun Kim*, Kyung Yong Ko, Hyungjun Kim, Jaehoon Kim, Jihun Oh, and Han-Bo-Ram Lee, Electronic Materials Letters, 12(3), 404-410 (2016), 5/10/2016, http://dx.doi.org/10.1007/s13391-016-5356-2, [Co-author]
88. "A Facile Method for Selective Healing of Graphene Defects Based on a Galvanic Displacement Reaction," Juree Hong, Jae-Bok Lee, Sanggeun Lee, Jungmok Seo, Hyunsoo Lee, Jeong Young Park, Tae-il Seo, Jong-Hyun Ahn, Taeyoon Lee*, and Han-Bo-Ram Lee*, NPG Asia Materials, 8, e262 (2016), 4/15/2016, http://dx.doi.org/10.1038/am.2016.42, [Corresponding author]
89. "Recent Advances in Atomic Layer Deposition," Neil P. Dasgupta, Stacey Bent, Paul S. Weiss, and Han-Bo-Ram Lee*, Chemistry of Materials, 28(7), 1943–1947 (2016), 4/12/2016, http://dx.doi.org/10.1021/acs.chemmater.6b00673, [Corresponding author]
90. "Formation of Ni Silicide from Atomic Layer Deposited Ni," Jaehong Yoon, Soo Hyeon Kim, Hangil Kim, Soohyun Kim, Hyungjun Kim*, and Han-Bo-Ram Lee*, Current Applied Physics, 16(7), 720-725 (2016), 4/6/2016, http://dx.doi.org/10.1016/j.cap.2016.04.005, [Corresponding author]
91. "Effects of Diffusion Barrier on Cu-gate ZnO:N Thin Film Transistors," Whang Je Woo, Taewook Nam, Hanearl Jung, Il-Kwon Oh, Han-Bo-Ram Lee, Wanjoo Maeng, and Hyungjun Kim*, IEEE Electron Device Letters, 37(5), 599-602 (2016), 3/31/2016, http://dx.doi.org/10.1109/LED.2016.2549035, [Co-author]
92. "Comparison of Hydrogen Sulfide Gas and Sulfur Powder for Synthesis of Molybdenum Disulfide Nanosheets," Jusang Park, Jeonggyu Song, Taejin Choi, Sangwan Sim, Hyunyong Choi, Sang Wook Han, Han-Bo-Ram Lee, Soo-Hyun Kim, and Hyungjun Kim*, Current Applied Physics, http://dx.doi.org/10.1016/j.cap.2016.03.022, 16(7), 691-695 (2016), 4/1/2016, [Co-author]
93. "Growth Characteristics and Electrical Properties of SiO_2 Thin Films Prepared using Plasma-Enhanced Atomic Layer Deposition and Chemical Vapor Deposition with an Aminosilane Precursor," Hanearl Jung, Woo-Hee Kim, Il-Kwon Oh, Chang-Wan Lee, Clement Lansalot-Matras, Su Jeong Lee, Jae-Min Myoung, Han-Bo-Ram Lee, and Hyungjun Kim*, Journal of Materials Science, 51(11), 5082-5091 (2016), 2/22/2016, http://dx.doi.org/10.1007/s10853-016-9811-0, [Co-author]
94. "Effects of Cl-based Ligand Structures on Atomic Layer Deposited HfO_2," Bo-Eun Park, Il-Kwon Oh, Chang Wan Lee, Gyeongho Lee, Young-Han Shin, Clement Lansalot-Matras, Wontae Noh,

Hyungjun Kim*, and Han-Bo-Ram Lee*, Journal of Physical Chemistry C, 120(11), 5958–5967 (2016), 3/11/2016, http://dx.doi.org/10.1021/acs.jpcc.5b05286, [Corresponding author]

95. "Plasma Enhanced Atomic Layer Deposition of SnO_2 Thin Films using $SnCl_4$ Precursor and Oxygen Plasma," Dong Kwon Lee, Zhixin Wan, Jong-Seong Bae, Han-Bo-Ram Lee, Ji-Hoon Ahn, Sang-Deok Kim, Jayong Kim, and Se-Hun Kwon*, Materials Letters, 166, 163-166 (2016), 3/1/2016, http://dx.doi.org/doi:10.1016/j.matlet.2015.12.049, [Co-author]

96. "Highly-conformal Nanocrystalline Molybdenum Nitride Thin Films by Atomic Layer Deposition using a Newly Synthesized Precursor as a Diffusion Barrier against Cu," Yujin Jang, Jun Beom Kim, Tae Eun Hong, So Jeong Yeo, Sunju Lee, Eun Ae Jung, Bo Keun Park, Taek-Mo Chung, Chang Kim, Do-Joong Lee, Han-Bo-Ram Lee, and Soo-Hyun Kim*, Journal of Alloys and Compounds, 663, 651-658 (2016), 4/5/2016, http://dx.doi.org/10.1016/j.jallcom.2015.12.148, [Co-author]

97. "A Controlled Growth of WN_x and WC_x Films Prepared by Atomic Layer Deposition," Jun Beom Kim, Byeong Hyeon Jang, Hyun-Jung Lee, Won Seok Han, Do-Joong Lee, Han-Bo-Ram Lee, Tae Eun Hong, and Soo-Hyun Kim*, Materials Letters, 168, 218-222 (2016), 4/1/2016, http://dx.doi.org/10.1016/j.matlet.2016.01.071, [Co-author]

98. "Wafer-scale, Conformal and Direct Growth of MoS_2 Thin Films by Atomic Layer Deposition," Yujin Jang, Seungmin Yeo, Han-Bo-Ram Lee, Hyungjun Kim, and Soo-Hyun Kim*, Applied Surface Science, 365, 160-165 (2016), 3/1/2016, http://dx.doi.org/10.1016/j.apsusc.2016.01.038, [Co-author]

99. "Complementary Unipolar WS_2 Field Effect Transistors using Fermi Level De-pinning Layers," Woojin Park, Yonghun Kim, Ukjin Jung, Jin Ho Yang, Chunhum Cho, Yun Ji Kim, Syed Mohammad Najib Hasan, Hyun Gu Kim, Han-Bo-Ram Lee, and Byoung Hun Lee*, Advanced Electronic Materials, 2(2), 1-5 (2016), 2/12/2016, http://dx.doi.org/10.1002/aelm.201500278, [Co-author]

100. "Self-Limiting Layer Synthesis of Transition Metal Dichalcogenides," Youngjun Kim, Jeong-Gyu Song, Yong Ju Park, Gyeong Hee Ryu, Su Jeong Lee, Jin Sung Kim, Pyo Jin Jeon, Chang Wan Lee, Whang Je Woo, Taejin Choi, Hanearl Jung, Han-Bo-Ram Lee, Jae-Min Myoung, Seongil Im, Zonghoon Lee, Jong-Hyun Ahn, Jusang Park, and Hyungjun Kim*, Scientific Reports, 6, 18754, 1/4/2016, http://dx.doi.org/10.1038/srep18754, [Co-author]

101. "Improved Corrosion Resistance and Mechanical Properties of CrN Hard Coatings with Atomic Layer Deposited Al_2O_3 Interlayer," Zhixin Wan, Teng Fei Zhang, Han-Bo-Ram Lee, Ji Hoon Yang, Woo Chang Choi, Byungchan Han, Kwang Ho Kim*, and Se-Hun Kwon*, ACS Applied Materials &

Interfaces, 7(48), 26716-26725 (2015), 11/23/2015, http://dx.doi.org/10.1021/acsami.5b08696, [Co-author]

102. "Formation of Continuous Pt Films on the Graphite Surface by Atomic Layer Deposition with Reactive O_3," Han-Bo-Ram Lee and Stacey F. Bent*, Chemistry of Materials, 27(19), 6802–6809 (2015), 10/1/2015, http://dx.doi.org/10.1021/acs.chemmater.5b03076, [First author]

103. "Area-Selective Chemical Vapor Deposition of Co for Cu Capping Layer," by Seung Wook Ryu, Soo Hyeon Kim, Jaehong Yoon, Jukka T. Tanskanen, Hyungjun Kim*, and Han-Bo-Ram Lee*, Current Applied Physics, 16(1), 88–92 (2016), 10/30/2015, http://dx.doi.org/10.1016/j.cap.2015.10.013, [Corresponding author]

104. "Nucleation and Growth of an HfO_2 Dielectric Layer for Graphene-Based Devices," Il-Kwon Oh, Jukka Tanskanen, Hanearl Jung, Kangsik Kim, Mi Jin Lee, Zonghoon Lee, Seung-Ki Lee, Jong-Hyun Ahn, Changwan Lee, Kwanpyo Kim, Hyungjun Kim*, and Han-Bo-Ram Lee*, Chemistry of Materials, 27 (17), 5868–5877 (2015), 8/19/2015, http://dx.doi.org/10.1021/acs.chemmater.5b01226, [Corresponding author]

105. "Reversible Liquid Adhesion Switching of Superamphiphobic Pd-decorated Ag dendrites via Gas-induced Structural Changes," Dayeong Kim, Jungmok Seo, Soonil Lee, Sera Shin, Hyeonjin Cho, Han-Bo-Ram Lee, and Taeyoon Lee*, Chemistry of Materials, 27 (14), 4964-4971 (2015), 6/30/2015, http://dx.doi.org/10.1021/acs.chemmater.5b01038, [Co-author]

106. "Influence of in situ Surface Cleaning on a Ge Substrate using TMA and $MgCp_2$ for HfO_2-based Gate Oxides", Il-Kwon Oh, Kangsik Kim, Zonghoon Lee, Jeong-Gyu Song, Chang- Wan Lee, David Thompson, Han-Bo-Ram Lee, Woo-Hee Kim, Wan Joo Maeng and Hyungjun Kim*, Journal of Materials Chemistry C, 3, 4852-4858 (2015), 4/22/2015, http://dx.doi.org/10.1039/C4TC02686A [Co-author]

107. "Internal and External Atomic Steps in Graphite Exhibit Dramatically Different Physical and Chemical Properties," Hyunsoo Lee, Han-Bo-Ram Lee, Sangku Kwon, Miquel Salmeron, and Jeong Young Park*, ACS Nano, 9(4), 3814-3819 (2015), 3/27/2015, http://dx.doi.org/10.1021/nn506755p [Co-author]

108. "One-step Hydrothermal Synthesis of Graphene Decorated V_2O_5 Nanobelts for Enhanced Electrochemical Energy Storage," Minoh Lee, Suresh Kannan Balasingam, Hu Young Jeong, Won G. Hong, Han-Bo-Ram Lee, Byung Hoon Kim*, and Yongseok Jun*, Scientific Reports, 5, 8151 (2015), 1/30/2015, http://dx.doi.org/10.1038/srep08151 [Co-author]

109. "Real-time Detection of Chlorine Gas using Ni/Si Shell/Core Nanowires," Dong-Jin Lee, Kwang Heo, Hyungwoo Lee, Joon-Hyung Jin, Hochan Chang, Minjun Park, Han-Bo-Ram Lee, Hyungjun

Kim, and Byung Yang Lee*, Nanoscale Research Letters, 10(1), 18 (2015), 1/28/2015, http://dx.doi.org/10.1186/s11671-015-0729-2, [Co-author]

110. "Plasma Enhanced Atomic Layer Deposition of Co on Metal Surfaces," Jae-Hong Yoon, Jeong-Gyu Song, Hyungjun Kim*, and Han-Bo-Ram Lee*, Surface and Coating Technology, 264, 60-65 (2015), 1/16/2015, http://dx.doi.org/10.1016/j.surfcoat.2015.01.019, [Corresponding author]
111. "Hydrophobicity of Rare Earth Oxides Grown by Atomic Layer Deposition," Il-Kwon Oh, Kang-Sik Kim, Zonghoon Lee, Kyung-Yong Ko, Su Jeong Lee, Jae-Min Myung, Clement Lansalot-Matras, Wontae Noh, Hyungjun Kim*, and Han-Bo-Ram Lee*, Chemistry of Materials, 27(1), 148-156 (2014), 12/19/2014, https://doi.org/10.1021/cm503659d, [Corresponding author]
112. "Selective Metal Deposition at Graphene Line Defects by Atomic Layer Deposition," Kwanpyo Kim† & Han-Bo-Ram Lee†, Richard W. Johnson, Jukka T. Tanskanen, Nan Liu, Myung-Gil Kim, Changhyun Pang, Chiyui Ahn, Stacey F. Bent, and Zhenan Bao*, Nature Communications, 5, 4781 (2014), 9/2/2014, https://doi.org/10.1038/ncomms5781, †Equal contribution, [First author]
113. "The Effect of O_3 on Growth of Pt by Atomic Layer Deposition," Han-Bo-Ram Lee, Katie L. Pickrahn, and Stacey F. Bent*, Journal of Physical Chemistry C, 118(23), 12325-12332 (2014), 6/2/2014, https://doi.org/10.1021/jp502596n, [First author]
114. "Vapor Transport Deposition and Epitaxy of Orthorhombic SnS on Glass and NaCl Substrates", Artit Wangperawong, Steven M. Herron, Rory R. Runser, Carl Hägglund, Jukka Tanskanen, Han-Bo-Ram Lee, Bruce M. Clemens, and Stacey F. Bent*, Applied Physics Letters, 103, 052105 (2013), 7/30/2013, http://dx.doi.org/10.1063/1.4816746, [Co-author]
115. "Self-assembly Based Plasmonic Arrays Tuned by Atomic Layer Deposition for Extremely Concentrated Visible Light Absorption," Carl Hägglund, Gabriel Zeltzer, Ricardo Ruiz, Isabell Thomann, Han-Bo-Ram Lee, Mark L. Brongersma, and Stacey F. Bent*, Nano Letters, 13(7), 3352-3357 (2013), 6/27/2013, http://dx.doi.org/10.1021/nl401641v, [Co-author]
116. "Plasma-enhanced Atomic Layer Deposition of Co using $Co(MeCp)_2$ Precursor", Jusang Park, Han-Bo-Ram Lee, Doyoung Kim, Jaehong Yoon, Clement Lansalot, Julien Gatineau, Henri Chevrel, and Hyungjun Kim*, Journal of Energy Chemistry, 22(3), 403-407 (2013), 5/1/2013, http://dx.doi.org/10.1016/S2095-4956(13)60052-2, [Co-author]
117. "Highly Sensitive, Patternable Organic Films at the Nanoscale Made by Bottom-up Assembly," Han Zhou, James M. Blackwell, Han-Bo-Ram Lee, and Stacey F. Bent*, ACS Applied Materials and Interfaces, 5(9), 3691-3696 (2013), 4/17/2013, http://dx.doi.org/10.1021/am4002887, [Co-author]

118. "Flexible Wireless Temperature Sensors Based on Ni Microparticle-Filled Binary Polymer Composites," Jin Jeon, Han-Bo-Ram Lee, and Zhenan Bao*, Advanced Materials, 25(6), 850-855 (2013), 2/13/13, http://dx.doi.org/10.1002/adma.201204082, [Co-author]

119. "Growth of Pt Nanowires by Atomic Layer Deposition on Highly Ordered Pyrolytic Graphite," Han-Bo-Ram Lee, Sung-Hyeon Baeck, Thomas F. Jaramillo, and Stacey F. Bent*, Nano Letters, 13(2), 457-463 (2013), 1/14/2013, http://dx.doi.org/10.1021/nl303803p, [First author]

120. "The Study of Ru Nanodot Synthesis using a CO_2 Supercritical Fluid Deposition," Doyoung Kim, Han-Bo-Ram Lee, Jaehong Yoon, and Hyungjun Kim*, Journal of Physics and Chemistry of Solids, 74(5), 664-667 (2013), 1/5/2013, http://dx.doi.org/10.1016/j.jpcs.2012.12.022, [Co-author]

121. "Nucleation-controlled Growth of Nanoparticles by Atomic Layer Deposition," Han-Bo-Ram Lee, Marja N. Mullings, Xirong Jiang, Bruce M. Clemens and Stacey F. Bent*, Chemistry of Materials, 24, 4051-4059 (2012), 10/19/2012, http://dx.doi.org/10.1021/cm3014978, [First author]

122. "The Low Temperature Atomic Layer Deposition of Ruthenium and the Effect of Oxygen Exposure," Rungthiwa Methaapanon, Scott M. Geyer, Han-Bo-Ram Lee, and Stacey F. Bent*, Journal of Materials Chemistry, 22, 25154-25156 (2012), 10/01/2012, http://dx.doi.org/10.1039/C2JM35332F, [Co-author]

123. "Active MnO_x Electrocatalysts Prepared by Atomic Layer Deposition for the Oxygen Evolution and Oxygen Reduction Reactions," Katie L. Pickrahn, Sang Wook Park, Yelena Gorlin, Han-Bo-Ram Lee, Thomas F. Jaramillo, and Stacey F. Bent*, Advanced Energy Materials, 2, 1269-1277 (2012), 06/14/12, http://dx.doi.org/10.1002/aenm.201200230, [Co-author]

124. "Initial Stage Growth during Plasma Enhanced Atomic Layer Deposition of Cobalt," Han-Bo-Ram Lee, Yong Jun Park, Sunggi Baik, and Hyungjun Kim*, Chemical Vapor Deposition, 18, 41-45 (2012), 03/05/12, http://dx.doi.org/10.1002/cvde.201106937, [First author]

125. "Silicidation of Co/Si Core Shell Nanowires," Han-Bo-Ram Lee, Gil Ho Gu, Chan Gyung Park, and Hyungjun Kim*, Journal of The Electrochemical Society, 159, K_{146}-K_{151} (2012), 03/02/12, http://dx.doi.org/10.1149/2.078205jes, [First author]

126. "Microstructure-Dependent Nucleation in Atomic Layer Deposition of Pt on TiO_2," Han-Bo-Ram Lee and Stacey F. Bent*, Chemistry of Materials, 24, 279-286 (2011), 12/29/11, http://dx.doi.org/10.1021/cm202764b, [First author]

127. "Supercritical Fluid Deposition of SiO_2 Thin Films: Growth Characteristics and Film Properties," Han-Bo-Ram Lee, In Chan Hwang, Jae-Min Kim, and Hyungjun Kim*, Journal of The Electrochemical Society, 159, D46-D49 (2011), 12/15/11, http://dx.doi.org/10.1149/2.031202jes, [First author]

128. “Atomic Layer Deposition of CdS Quantum Dots for Solid-State Quantum Dot Sensitized Solar Cells,” Thomas P. Brennan, Pendar Ardalan, Han-Bo-Ram Lee, Jonathan R. Bakke, I-Kang Ding, Michael D. McGehee, Stacey F. Bent*, Advanced Energy Materials, 1, 1169-1175 (2011), 10/04/11, http://dx.doi.org/10.1002/aenm.201100363, [Co-author]

129. “Atomic Layer Deposition of Co using N_2/H_2 Plasma as a Reactant,” Jaehong Yoon, Han-Bo-Ram Lee, Doyoung Kim, Taehoon Cheon, Soo-Hyun Kim, and Hyungjun Kim*, Journal of The Electrochemical Society, 158, H_{1179}-H_{1182} (2011), 10/05/11, http://dx.doi.org/10.1149/2.077111jes, [Co-author]

130. “In-situ Synchrotron X-ray Scattering Study of Thin Film Growth by Atomic Layer Deposition,” Yong Jun Park, Dong Ryeol Lee, Hyun Hwi Lee, Han-Bo-Ram Lee, Hyungjun Kim, Gye-Choon Park, Shi-Woo Rhee, and Sunggi Baik*, Journal of Nanoscience and Nanotechnology, 11, 1577-1580 (2011), 02/01/11, http://dx.doi.org/10.1166/jnn.2011.3399, [Co-author]

131. “Effects of Self-Assembled Monolayers on Solid- State CdS Quantum Dot Sensitized Solar Cells,” Pendar Ardalan, Thomas P. Brennan, Han-Bo-Ram Lee, Jonathan R. Bakke, I-Kang Ding, Michael D. McGehee, Stacey F. Bent*, ACS Nano, 5, 1495-1504 (2011), 02/07/11, http://dx.doi.org/10.1021/nn103371v, [Co-author]

132. “Atomic Layer Deposition of Ni Thin Films and Application to Area-Selective Deposition,” Woo-Hee Kim, Han-Bo-Ram Lee, Kwang Heo, Young Kuk Lee, Taek-Mo Chung, Chang Gyoun Kim, Seunghun Hong, Jong Heo, and Hyungjun Kim*, Journal of The Electrochemical Society, 158, D1-D5 (2011), 11/09/10, http://dx.doi.org/10.1149/1.3504196, [Co-author]

133. “Area Selective Atomic Layer Deposition by Microcontact Printing with a Water-Soluble Polymer,” Marja N. Mullings, Han-Bo-Ram Lee, Nathan Marchack, Xirong Jiang, Zhebo Chen, Yelena Gorlin, Kuang-Po Lin, and Stacey F. Bent*, Journal of The Electrochemical Society, 157, D600-D604 (2010), 10/04/10, http://dx.doi.org/10.1149/1.3491376, [Co-author]

134. “Deposition of Ultra-Thin Polythiourea Films by Molecular Layer Deposition,” Paul W. Loscutoff, Han-Bo-Ram Lee, and Stacey F. Bent*, Chemistry of Materials, 22, 5563-5569 (2010), 09/08/10, http://dx.doi.org/10.1021/cm1016239, [Co-author]

135. “Self-formation of Dielectric Layer Containing $CoSi_2$ Nanocrystals by Plasma-Enhanced Atomic Layer Deposition,” Han-Bo-Ram Lee and Hyungjun Kim*, Journal of Crystal Growth, 312, 2215-2219 (2010), 05/05/10, http://dx.doi.org/10.1016/j.jcrysgro.2010.04.048, [First author]

136. “Plasma-Enhanced Atomic Layer Deposition of Ni,” Han-Bo-Ram Lee, Sung-Hwan Bang, Woo-Hee Kim, Gil Ho Gu, Young Kuk Lee, Taek-Mo Chung, Chang Gyoun Kim, C. G. Park, and Hyungjun Kim, Japanese Journal of Applied Physics, 49, 05FA11 (2010), 05/20/10, http://dx.doi.org/10.1143/JJAP.49.05FA11, [First author]

137. "Plasma-Enhanced Atomic Layer Deposition of Cobalt Using Cyclopentadienyl Isopropyl Acetamidinato-Cobalt as a Precursor," Jae-Min Kim, Han-Bo-Ram Lee, Clement Lansalot, Christian Dussarrat, Julien Gatineau, and Hyungjun Kim*, Japanese Journal of Applied Physics, 49, 05FA10 (2010), 05/20/10, http://dx.doi.org/10.1143/JJAP.49.05FA10, [Co-author]

138. "Degradation of Deposition Blocking Layer during Area Selective Plasma Enhanced Atomic Layer Deposition of Cobalt," Han-Bo-Ram Lee, Woo-Hee Kim, Jeong Won Lee, Jaemin Kim, Inchan Hwang, and Hyungjun Kim*, Journal of Korean Physical Society, 56, 104-107 (2010), 12/01/09, http://dx.doi.org/10.3938/jkps.56.104, [First author]

139. "High Quality Area-Selective Atomic Layer Deposition Co using Ammonia Gas as a Reactant," Han-Bo-Ram Lee, Woo-Hee Kim, Jeong Won Lee, Kwang Heo, In Chan Hwang, Yongjun Park, Seunghun Hong, and Hyungjun Kim*, Journal of The Electrochemical Society, 157, D10-D15 (2010), 11/03/09, http://dx.doi.org/10.1149/1.3248002, [First author]

140. "The Supercritical Fluid Deposition of Conformal $SrTiO_3$ Films with Composition Uniformity in Nano-contact Holes," J. H. Lee, J. Y. Son, Han-Bo-Ram Lee, Heung-Soon Lee, D.-J. Ma, C.-S. Lee, and Hyungjun Kim*, Electrochemical and Solid State Letters, 12, D45-D47 (2009), 03/06/09, http://dx.doi.org/10.1149/1.3092891, [Co-author]

141. (Review Article) "Applications of Atomic Layer Deposition to Nanofabrication and Emerging Nanodevices," Hyungjun Kim*, Han-Bo-Ram Lee, and W.-J. Maeng, Thin Solid Films, 517, 2563-2580 (2009), 02/27/09, http://dx.doi.org/10.1016/j.tsf.2008.09.007, [Co-author]

142. "Spontaneous Formation of Vertical Magnetic Metal Nanorod Arrays during Plasma Enhanced Atomic Layer Deposition," Han-Bo-Ram Lee, Gil Ho Gu, J. Y. Son, C.G. Park, and Hyungjun Kim*, Small, 4, 2247-2254 (2008), 10/30/08, http://dx.doi.org/10.1002/smll.200801074, [First author]

143. "Thermal and Plasma Enhanced Atomic Layer Deposition Ruthenium and Electrical Characterization as a Metal Electrode," Sang-Joon Park , Woo-Hee Kim, Han-Bo-Ram Lee, W. J. Maeng, and Hyungjun Kim*, Microelectronic Engineering, 85, 39-44 (2008), 02/12/07, http://dx.doi.org/10.1016/j.mee.2007.01.239, [Co-author]

144. "High Quality Epitaxial $CoSi_2$ using Plasma Nitridation-Mediated Epitaxy; The Effects of Capping Layer," Han-Bo-Ram Lee, Gil Ho Gu, J.Y. Son, C.G. Park, and Hyungjun Kim*, Journal of Applied Physics, 102, 094509 (2007), 11/08/07, http://dx.doi.org/10.1063/1.2805649, [First author]

145. "Stress Dependence of Growth Mode Change of Epitaxial Layered Cobaltite γ-$Na_{0.7}CoO_2$," J. Y. Son, Han-Bo-Ram Lee, and J. H. Cho*, Applied Surface Science, 254, 436-440 (2007), 06/06/07, http://dx.doi.org/10.1016/j.apsusc.2007.05.074, [Co-author]

146. "Surface Charge on Ferroelectric Thin Film by The High Electric Field Induced at Scanning Probe Microscope Tip," J. Y. Son, Young-Han Shin, Han-Bo-Ram Lee, Hyungjun Kim, J. H. Cho*, and A. I. Ali*, Journal of Korean Physical Society, 51, S_{125}-S_{128} (2007), 10/31/07, [Co-author]
147. "Nitride Mediated Epitaxy of $CoSi_2$ Through Self-interlayer-formation of Plasma-Enhanced Atomic Layer Deposition Co," Han-Bo-Ram Lee, J. Y. Son and Hyungjun Kim*, Applied Physics Letters, 90, 123509 (2007), 03/23/07, http://dx.doi.org/10.1063/1.2742791, [First author]
148. "High Quality Cobalt Thin Films by Plasma-Enhanced Atomic Layer Deposition," Han-Bo-Ram Lee and Hyungjun Kim*, Electrochemical and Solid State Letters, 9, G323-G325 (2006), 08/29/06, http://dx.doi.org/10.1149/1.2338777, [First author]

KOREAN (NRF)

1. "Growth of Atomic Layer Deposition Platinum on TiO_2," Hyun Gu Kim and Han-Bo-Ram Lee*, 48 (2), 38-42, Journal of the Korean Institute of Surface Engineering, http://dx.doi.org/10.5695/JKISE.2015.48.2.38, [Corresponding author]
2. "원자층증착기술에서 표면특성에 따른 나노재료의 차원변환기술," Han-Bo-Ram Lee, Journal of Informative Display, 14(5), 23-32 (2013), 10/31/2013, http://www.k-ids.or.kr/search/paper_list.asp, [First author]

Go to the 1st page

Patents

GRANTED

1. Woojin Lee, Han-Bo-Ram Lee, Sanghoon Ahn, Ik Su Kim, Sumaira Yasmeen, Byung Kook Ko, Bonwook Gu, “Method for Fabricating Semiconductor Device,” 10-3015198, 9/1/2026 (with Samsung Electronics)
2. Han-Bo-Ram Lee, Do Han Lee, Eun Soo Kim, Seung Wook Ryu, Tae Hwan Lim, Abu Saad Aqueel Ahmad Ansari, Ngoc Le Trinh, “Thin Film Deposition Method and Method of Manufacturing Electronic Device Using the Same,” 10-2981295, 6/18/2026
3. Han-Bo-Ram Lee, Do Han Lee, Eun Soo Kim, Seung Wook Ryu, Tae Hwan Lim, Abu Saad Aqueel Ahmad Ansari, Ngoc Le Trinh, “Thin Film Deposition Method and Method of Manufacturing Electronic Device Using the Same,” US 12,604,681, 4/14/2026
4. Han-Bo-Ram Lee, Do Han Lee, Eun Soo Kim, Seung Wook Ryu, Tae Hwan Lim, Abu Saad Aqueel Ahmad Ansari, Ngoc Le Trinh, “Thin Film Deposition Method and Method of Manufacturing Electronic Device Using the Same,” US 12,540,395, 2/3/2026
5. Han-Bo-Ram Lee, Summal Zoha, “Selective Deposition Method of Thin Film Using Area-selective Atomic Layer Deposition and Substrate on Which a Thin Film is Selectively Formed,” KR 10-2892382, 11/25/2025
6. Han-Bo-Ram Lee, Chan Hee Mun, “Methods for Implementation of Icephobic Surface with Polymer and Micro/Nano Powder,” 10-2489091, 1/11/2023
7. Han-Bo-Ram Lee, Chi Thang Nguyen, “Method for Fabricating a Thin Film and a Thin Film Fabricated by the Same,” 10-2341035, 12/15/2021
8. Han-Bo-Ram Lee, Woo Hee Kim, “A Defect Healing Method by Selective Deposition for Memory Device,” 10-2225183, 3/3/2021
9. Han-Bo-Ram Lee, Hyun Gu Kim, Woo Hyeok Kwon, “A Color Electronic Textile and Preparation Method Thereof,” 10-2172190, 10/26/2020
10. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, “Superhydrophobic Coating Material and Method for Manufacturing the Same,” US 10,392,701 B2, 08/27/2019
11. Han-Bo-Ram Lee, Jae-Kwang Lee, Mohammed Rizwan Khan, “Method for Implementing Hydrophobic Surface using Spin Coating,” KR 10-1974955, 4/26/2019
12. Han-Bo-Ram Lee, Jae-Kwang Lee, Mohammed Rizwan Khan, “Method for Implementing Hydrophobic Surface using Spray Coating,” 10-1974953, 4/26/2019

13. Han-Bo-Ram Lee, Hyungjun Kim, Chang Mo Yoon, Il-Kwon Oh, "Method for Forming Thin Film," KR 10-1942819, 1/22/2019
14. Han-Bo-Ram Lee, "Fabrication Method of High Quality Materials for Quadruple Patterning using Heteroatom Alloying," KR 10-1900181, 9/12/2018
15. Han-Bo-Ram Lee, "Method for Forming Superhydrophobic Coating Layer using PDMS and Micropowders," KR 10-1871073, 6/19/2018
16. Han-Bo-Ram Lee, "Method for Modifying Hydrophobic Surfaces using Hydrophobic Ceramic Powders Having Core-shell Structure," KR 10-1860206, 5/15/2018
17. Han-Bo-Ram Lee, "Fabrication Method of Conductive Fibers," KR 10-1846084, 3/30/2018
18. Han-Bo-Ram Lee, Jaehong Yun, "Method for Forming Hydrophobic Coating Layer using Structuralizaion of PDMS Surface," KR 10-1807407, 12/4/2017
19. Han-Bo-Ram Lee, Hyun Gu Kim, "Self-healing Method of Self-healing Polymer using Defect-healed Reduced Graphene Oxide Heater," KR 10-1807459, 12/4/2017
20. Han-Bo-Ram Lee, Hyungjun Kim, Bo-Eun Park, Il-Kwon Oh, "Enhanced Electric Device for MOS Capacitor and the Manufacturing Method thereof," KR 10-1752059, 6/22/2017
21. Han-Bo-Ram Lee, Hyungjun Kim, Bo-Eun Park, Il-Kwon Oh, "Enhanced electrical characteristic Electric Device for MIM Capacitor and the Manufacturing Method thereof," KR 10-1752060, 6/22/2017
22. Han-Bo-Ram Lee, Hyungjun Kim, Jun Ho Choi, Il-Kwon Oh, "Method for Modifying Surface of Substrate Using Rare Earth Oxide Thin Film," KR 10-1751619, 6/21/2017
23. Taeyoon Lee, Jong Hyun Ahn, Juree Hong, Jae-Bok Lee, Han-Bo-Ram Lee, "Method for Healing Defect of Conductive Layer, Method for Forming Metal-carbon Compound Layer, 2D Nano Materials, Transparent Electrode and Method for Manufacturing the Same," KR 10-1720168, 3/21/2017
24. Han-Bo-Ram Lee, "3D Conductive Coating Method For Clear SEM Measurement," KR 10-1686386, 12/7/2016
25. Hyungjun Kim, Il-Kwon Oh, Chang-Mo Yun, Han-Bo-Ram Lee, "Filter For Removing Moisture And Method For Manufacturing The Same," KR 10-1670337, 10/24/2016
26. Han-Bo-Ram Lee, Hyun Gu Kim, "Manufacturing Method of Ultrathin Continuous Metal Film Using Surface Funtionalization," KR 10-1672984, 10/31/2016

27. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, "Apparatus for Plasma Enhanced Atomic Layer Deposition and Method for Forming Thin Film Oxides Using The Same," KR 10-1662194, 9/27/2016
28. Hyungjun Kim, Soo Hyun Kim, Jaehong Yun, Han-Bo-Ram Lee, "Semiconductor Device and Method For Manufacturing the Same," US 9,396,994, 7/19/2016
29. Hyungjun Kim, Soo Hyun Kim, Jaehong Yun, Han-Bo-Ram Lee, "Semiconductor Device and Method For Manufacturing the Same," KR 10-1621852, 5/11/2016
30. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, "Superhydrophobic Coating Material and Method for Manufacturing the Same," KR 10-1617396, 4/26/2016
31. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, "Method For Forming Coating Layer and Coating Material Having Waterproof Property," KR 10-1615897, 4/21/2016
32. Han-Bo-Ram Lee, "Manufacturing Method of Nanowire using Atomic Layer Deposition," KR 10-1602628, 3/7/2016
33. Gilsang Yoo, Il-Kwon Oh, Hyungjun Kim, Han-Bo-Ram Lee, "Apparatus Method for Performing Plasma Enhanced Atomic Layer Deposition Employing Very High Frequency," KR 10-1596329, 2/16/2016
34. Han-Bo-Ram Lee and Kwan Pyo Kim, "Healing Method of Defect using Atomic Layer Deposition," KR 10-1568159, 11/5/2015
35. Jaehong Yun, Soo Hyun Kim, Hyungjun Kim, Han-Bo-Ram Lee, "Copper Layer Capping Method and Method for Fabricating Copper Wire Employing the Same," KR 10-1553357, 9/9/2015
36. Han-Bo-Ram Lee, Jaemin Kim, Jeong Won Lee, Heung-Soon Lee, Hyungjun Kim, "Method for Fabrication of Metal Nanostructures by using Supercritical Fluid Deposition," KR 10-1124351, 2/29/2012
37. Han-Bo-Ram Lee, Gil Ho Gu, C.G. Park, Hyungjun Kim, "Manufacturing Method of Metal Nanostructures," KR 10-1124352, 2/29/2012
38. Han-Bo-Ram Lee, Woo-Hee Kim, Hyungjun Kim, "Method for Forming Contacts of Semiconductor Devices using The Selective Deposition," KR 10-1078309, 10/25/2011
39. Han-Bo-Ram Lee, Gil Ho Gu, C.G. Park, J.Y. Son, Hyungjun Kim, "Fabrication Method of Catalyst-less Metal Nanorods by Plasma-Enhanced Atomic Layer Deposition and A Semiconductor Element," KR 10-0920456, 9/29/2009
40. Han-Bo-Ram Lee, J.Y. Son, Hyungjun Kim, "Manufacturing Method of Metal Silicide Thin Layer by Plasma-Enhanced Atomic Layer Deposition without Annealing," KR 10-0920455, 9/29/2009

41. Han-Bo-Ram Lee, J.Y. Son, Hyungjun Kim, “Manufacturing Method of Metal Silicide by Plasma-Enhanced Atomic Layer Deposition for Contact Application in Semiconductor Devices,” KR 10-0872799, 12/2/2008
42. Han-Bo-Ram Lee, Gil Ho Gu, C.G. Park, Hyungjun Kim, “Method for Forming Metal-Silicide Layer in Semiconductor Devices using Plasma Nitridation,” KR 10-0872801, 12/2/2008
43. Hyungjun Kim, Y.H. Shin, Y.S. Woo, Han-Bo-Ram Lee, J.Y. Son, “Transistor and Nonvolatile Memory using Deformation Resistivity of Carbon Nanotube and Piezoelectric Effect,” KR 10-0848813, 7/28/2007

PENDING

1. Han-Bo-Ram Lee, Hyewon Park, “Atomic Layer Deposition Method and Semiconductor Device using the Same,” 10-2025-0029862, 3/7/2025
2. Han-Bo-Ram Lee, Summal Zoha, Yongjin Kim, Sanghoon Ahn, Minkyung Lee, Woojin Lee, Hangyeol Choi, “Semiconductor Devices,” 10-2024-0087160, 7/2/2024
3. Han-Bo-Ram Lee, Il-Kwon Oh, “Dysprosium as a Crystalline Aligner in HfO_2 Thin Films by Atomic Layer Deposition,” 10-2022-0164227, 11/30/2022
4. Han-Bo-Ram Lee, Summal Zoha, “Area Selective Deposition of HfO_2 ALD on Metal, Oxide, and Nitride Using Organothiol Inhibitor,” 10-2022-0047877, 4/19/2022
5. Han-Bo-Ram Lee, Do Han Lee, Eun Soo Kim, Seung Wook Ryu, Sung Hwan Jo, Abu Saad Aqueel Ahmad Ansari, Ngoc Le Trinh, “Thin Film Deposition Method And Manufacturing Method Of Electronic Device Applying The Same,” 10-2022-0021554, 2/18/2022
6. Taekjib Choi, Ho-Jin Lee, Jun-Bong Lee, Han-Bo-Ram Lee, Hong-Woon Lee, Byung-Kook Ko, Jae Kwang Lee, “Self-rectifying Ferroelectric Tunnel Junction Memory Device and Crosspoint Array Having the Same,” 10-2018-0037603, 3/30/2018
7. Han-Bo-Ram Lee, “Fabrication Method of High Quality Materials for Quadruple Patterning using Heteroatom Alloying,” PCT/KR2017/013109, 11/17/2017
8. Hyungjun Kim, Il-Kwon Oh, Chang-Mo Yun, Han-Bo-Ram Lee, “Method for Forming Thin Film,” PCT/ KR2016/014333, 12/7/2016
9. Hyungjun Kim, Il-Kwon Oh, Chang-Mo Yun, Han-Bo-Ram Lee, “Filter For Removing Moisture And Method For Manufacturing The Same,” PCT/KR2016/014160, 12/2/2016
10. Taeyoon Lee, Jong Hyun Ahn, Juree Hong, Jae-Bok Lee, Han-Bo-Ram Lee, “Method for Healing Defect of Conductive Layer, Method for Forming Metal-Carbon Compound Layer, 2D Nano

Materials, Transparent Electrode and Method for Manufacturing the Same," US 15/196,820, 06/29/2016

11. Han-Bo-Ram Lee, Hyungjun Kim, Chang Mo Yoon, Il-Kwon Oh, "A Functional Filter and Manufacturing Method of The Same," KR 10-2015-0171229, 12/3/2015
12. Hyungjun Kim, Il-Kwon Oh, Han-Bo-Ram Lee, "Method for Forming Coating Layer and Coating Material Having Waterproof Property," US 14/814,888, 07/31/2015
13. Jaehong Yun, Han-Bo-Ram Lee, Hyungjun Kim, "Method for Fabrication of Silicide Nanowires with Ru Capping Layer," KR 10-2010-0140092, 12/31/2010

Go to the 1st page

Technology Transfer

1. "Fabrication Method of Conductive Fibers," KR 10-1846084, transferred to Hanil Corporation, 5/30/2025
2. "Method for Forming Superhydrophobic Coating Layer using PDMS and Micropowders," KR 10-1871073, transferred to Pyocell Co., Ltd., 9/23/2024
3. "Method for Forming Hydrophobic Coating Layer using Structuralization of PDMS Surface," KR 10-1807407, transferred to H-Steel Co., Ltd., 8/9/2024
4. "Manufacturing Method of Ultrathin Continuous Metal Film Using Surface Functionalization," KR 10-1672984, transferred to Moon-Ju Park, 1/4/2024
5. "Healing Method of Defect using Atomic Layer Deposition," KR 10-1568159, transferred to ADWIN Co., Ltd., 11/15/2023
6. "Manufacturing Method of Nanowire using Atomic Layer Deposition," KR 10-1602628, transferred to Daon F&E Co., Ltd., 4/7/2023
7. "3D Conductive Coating Method For Clear SEM Measurement," KR 10-1686386, transferred to Moman Co., Ltd., 8/30/2018

Go to the 1st page

Book Chapters & Proceedings

BOOK CHAPTERS

1. Summal Zoha, Byungchan Lee, and Han-Bo-Ram Lee*, "Defect Chemistry and Associated Organic Functionalization Strategies in Area-Selective Deposition," in the section of "Organic Functionalization of Inorganic Surafces," submitted to the Encyclopedia of Interfacial Chemistry, 2nd Edition, Elsevier. [Corresponding author]
2. Hyungjun Kim, Soo Hyun Kim, and Han-Bo-Ram Lee, "Back End of the Line," in Atomic Layer Deposition for Semiconductors, Ed. Cheol Seong Hwang, Springer, ISBN: 978-1-4614-8054-9, 10/19/2013, http://dx.doi.org/10.1007/978-1-4614-8054-9_8, [Co-author]
3. Han-Bo-Ram Lee and Stacey F. Bent, "Chapter 9. Nanopatterning by Area-Selective Atomic Layer Deposition," in Atomic Layer Deposition of Nanostructured Materials, Ed. Nicola Pinna and Mato Knez, Wiley-VCH (2011), ISBN: 978-3-527-32797-3, 1/2/2012, http://dx.doi.org/10.1002/9783527639915.ch9, [First author]

PROCEEDINGS

1. "Contact Resistance Reduction using Fermi Level De-pinning Layer for MoS_2 FETs," Woojin Park, Yonghun Kim, Sang Kyung Lee, Ukjin Jung, Jin Ho Yang, Chunhum Cho, Yun Ji Kim, Sung Kwan Lim, In Seol Hwang, Han-Bo-Ram Lee, and Byoung Hun Lee, International Electron Device Meeting, 5.1.1 - 5.1.4 (2014), http://dx.doi.org/10.1109/IEDM.2014.7046986, [Co-author]
2. Hyungjun Kim, Jaehong Yoon, and Han-Bo-Ram Lee, "Atomic layer deposition for nanoscale contact applications," Interconnect Technology Conference and 2001 Materials for Advanced Metallization (IITC/ MAM), 2011 IEEE International, 8-12, 1-3(2011), http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=5940260, [Co-author]
3. Hyungjun Kim, Han-Bo-Ram Lee, Woo-Hee Kim, Sang-Joon Park, and In Chan Hwang, "Nanomaterials Fabrication using Advanced Thin Film Deposition and Nanohybrid Process," IEEE Nanotechnology Materials and Devices Conference 2009, 5167578, 3-4(2009), http://ieeexplore.ieee.org/xpls/abs_all.jsp?arnumber=5167578, [Co-author]
4. Hyungjun Kim, Woo-Hee Kim, Han-Bo-Ram Lee, and S. J. Lim, "The Benefits of Atomic Layer Deposition in Non-Semiconductor Applications; Producing Metallic Nanomaterials and Fabrication of Flexible Display," ECS Transactions, 25, 101(2009), http://dx.doi.org/10.1149/1.3205047, [Co-author]

5. Han-Bo-Ram Lee, Kwang Heo, Seunghun Hong, and Hyungjun Kim, “Formation of Silicide Nanowires by Annealing of Atomic Layer Deposition Cobalt/Silicon Core-Shell Nanowires,” ECS Transactions, 25, 175(2009), http://dx.doi.org/10.1149/1.3205052, [First author]
6. (Invited) Han-Bo-Ram Lee and Hyungjun Kim, “Atomic Layer Deposition for Nanoscale Contact Applications,” 25th International VLSI/ULSI Multilevel Interconnection Conference Proceeding, 39(2008), [First author]
7. Han-Bo-Ram Lee and Hyungjun Kim, “Area Selective Atomic Layer Deposition of Cobalt Thin Films”, ECS Transactions, 16, 219(2008), http://dx.doi.org/10.1149/1.2979997, [First author]

Go to the 1st page

Invited Conference Presentations

1. "Atomic Hydrogen for Oxygen-Free Metal ALD and Bottom-Up Interconnect Fill," ADMETA 2026, Japan, hybrid, October 7-9, 2026
2. "Current and Future Perspectives of Area-Selective Deposition," ALD/ALE2026, Tampa, FL, USA, June 28, 2026
3. "Surface Chemistry to Device Innovation: Expanding the Frontiers of Atomic Layer Deposition," 7th International Conference on Applied Surface Science, Chengdu, China, June 1, 2026
4. "Atomic Layer Deposition in the Age of AI: A Reinforcing Loop from Angstroms to Algorithms," 2nd Chem & Bio Engineering Forum, Hangzhou, China, May 7, 2026
5. "Don't Cover Everything: Why Semiconductor Processes Learned to Be Selective (다 덮지 마세요: 반도체 공정 이 선택을 배운 이유)," KALD 2026, Seoul, Republic of Korea, April 9, 2026
6. "New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication," Japanese Society of Applied Physics Spring Meeting, Tokyo, Japan, March 15, 2026
7. "New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication," RSC and EMRC Joint Symposium for Advancing Materials Science, Doha, Qatar, November 3, 2025
8. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," IMRC, Cancun, Mexico, August 18, 2025
9. "Mastering Selective Deposition in One Hour: Principles and Practice (선택적 증착 1시간만에 뽀개기 원리와 실 전)," KALD 2025, Seoul, Republic of Korea, April 24, 2025
10. "Building 3D AI Semiconductor Devices from the Ground Up: The Role of Atomic Layer Deposition (원자층 증 착법으로 부터 시작되는 3차원 AI 반도체 소자 공정기술)," AI Workshop, Korean Society of Electronic and Electrical Engineering, Seoul, Republic of Korea, April 17, 2025
11. "Area Selective Deposition Fundamentals & Applications, Semicon Korea, Seoul, Republic of Korea, February 19, 2025
12. "Rediscovery of ALD to Overcome the Limitations of Semiconductor Manufacturing," Asian Pacific ALD 2024, Shanghai, China, October 17, 2024
13. "Technological Expansion of Atomic Layer Deposition to Patterning beyond Thin Film Deposition," ACS 2024 Fall, Denver, CO, USA, August 19, 2024
14. "Area Selective Atomic Layer Deposition using a Size Cutter," ALD/ALE2024, Helsinki, Finland, August 5, 2024

15. "Area Selective ALD using Homometallic Precursor Inhibitors for High Process Compatibility," ASD Workshop 2024, Montreal, Canada, April 15, 2024
16. "Atomic Layer Deposition & Selective Growth for Nanofabrication of the Si Chips," ICAE 2023, Jeju, Republic of Korea, November 2, 2023
17. "Area Selective Deposition using Homometallic Precursor Inhibitors," ECS244, Gothenburg, Sweden, October 9, 2023
18. "Area Selective Deposition; The 4th Toolbox for Nanofabrication of Si Devices," Nano Korea 2023, Goyang, Republic of Korea, July 6, 2023
19. "Computational Modeling of Physical Surface Reactions of Precursors in Atomic Layer Deposition by Monte Carlo Simulations," APATCC-10, Quy Nhon, Vietnam, February 20, 2023
20. "Seam-less Deposition of TiO_2 using Gradient Selective Deposition," KISM 2022, Pusan, Republic of Korea, November 4, 2022
21. "The 4th Toolbox for Semiconductor Device Nanofabrication beyond ALD," Carbon Symposium, Seoul, Republic of Korea, September 23, 2022
22. "Another Opportunity in AS-ALD using Precursor Inhibitors," ALD/ALE 2021, online, June 28, 2021
23. "Atomic Layer Deposition Fundamentals & Applications," ICMAP, online, January 20, 2021
24. "Surface Energy Engineering by High & Low Technologies," The 4th International Conference on Active Materials and Soft Mechatronics, Songdo, Republic of Korea, October 16 - 19, 2019
25. "Back to the Surface Science Again," China ALD Workshop 2019, Huangshuan, China, August 8 - 9, 2019
26. "Surface-reactivity-determined Patterning Technology; The Era of Atomic Crafting," ASD Workshop 2019, Leuven, Belgium, April 4 - 5, 2019
27. "Surface-reactivity-determined Patterning Technology; A Route to Atomic Craft," Nano Convergence Conference 2019, Gangchon, Republic of Korea, January 17 - 18, 2019
28. "Surface Energy Tuning in Atomic Scale for Hydrophobic Coating," The 4th International Conference on ALD Applications & 2018 China ALD Conference, Shenzen, China, October 14 - 17, 2018
29. "Atomic Layer Deposition of Rare Earth Oxides for Surface Energy Control," The 7th International Conference on Microelectronics and Plasma Technology, Songdo, Republic of Korea, July 24 - 28, 2018

30. "Surface Functionalization for Conductivity Improvement by Metal Atomic Layer Deposition," IEEE International Interconnect Technology Conference 2018, Santa Clara, CA, June 4 - 7, 2018
31. "Surface Energy Tuning in Atomic Scale for Hydrophobic Coating," The Korean Ceramic Society Spring Meeting 2018, Changwon, Republic of Korea, April 11 - 13, 2018
32. "Atomic Layer Deposition on T-shirts," American Chemical Society 2018 Spring National Meeting, New Orleans, LA, March 18 - 22, 2018
33. "Atomic Layer Deposited Metal Thin Films on T-shirts," Korean Institute of Surface Engineering Meeting 2017, Jeju, Republic of Korea, November 22 - 25, 2017
34. "Atomic Layer Deposited Metal Thin Films on T-shirts," Korean Vacuum Society Summer Meeting 2017, Hongcheon, Republic of Korea, August 16 - 18, 2017
35. "Defect Healing of Graphene for Conductive Flexible Electrodes by Wet & Dry Methods," International Conference on Electronic Materials and Nanotechnology for Green Environment (ENGE) 2016, Jeju, Republic of Korea, November 6 - 9, 2016
36. "Atomic Layer Deposition of Metals on Textiles for Wearable Electronics," China ALD 2016, Suzhou, China, October 16 - 19, 2016
37. "Electronic Textiles Fabricated using Atomic Layer Deposition," CIMTEC 2016, Perugia, Italy, June 5 - 10, 2016
38. "Textile-Based Pressure Sensor Prepared by Atomic Layer Deposition," Korean Society of Opticalelectronics Meeting 2016, Gwangju, Republic of Korea, January 26 - 27, 2016
39. "Atomic Layer Deposition of Rare Earth Oxide for Hydrophobic Coating," Korean Institute of Surface Engineering Fall Meeting 2015, Yongin, Republic of Korea, November 26 - 27, 2015
40. "Opportunities of Atomic Layer Deposition for Electronic Textile Applications," The Korean Fiber Society Fall Meeting 2015, Busan, Republic of Korea, November 5 - 6, 2015
41. "Opportunities in Atomic Layer Deposition for Electronic Textile and Hydrophobic Coating Applications," 228th Electrochemical Society Meeting, Phoenix, AZ, USA, October 11 - 15, 2015
42. "ALD of Noble Metals for Energy Application," 14th International Conference on Atomic Layer Deposition, Kyoto, Japan, June 15 - 18, 2014
43. "Pt Nanoparticle on Site-Constrained Surface by Atomic Layer Deposition," The 6th Asian Conference on Crystal Growth and Crystal Technology, Jeju, Republic of Korea, June 11 - 14, 2014
44. "Pt Nanoparticle Fabrication by Atomic Layer Deposition with Self-Assembled Monolayers," The Korean Ceramic Society Fall Meeting 2013, Jeju, Republic of Korea, October 16 - 18, 2013

45. "Nucleation Control in Atomic Layer Deposition by using Self-Assembled Monolayer," Korean Materials Research Society Spring Meeting 2013, Yeosu, Republic of Korea, May 23-24, 2013
46. "Nucleation & Growth of ALD Metal on Carbon Surfaces," Korean Atomic Layer Deposition Workshop 2013, Seoul, Republic of Korea, April 26, 2013
47. "Nucleation of 1D Pt Nanowires by Atomic Layer Deposition on Highly Ordered Pyrolytic Graphite," 2013 Spring Conference of the Korean Institute of Metals & Materials, Jeju, Republic of Korea, April 25 - 26, 2013
48. "Atomic Layer Deposition of Metals on 2D Materials," China Semiconductor Technology International Conference 2013, Shanghai, China, March 16 - 18, 2013
49. "Atomic Layer Deposition for Nanoscale Contact Applications," International Conference on Electronic Materials and Nanotechnology for Green Environment, Jeju, Republic of Korea, November 21 - 24, 2010
50. "The Applications of Atomic Layer Deposition of Cobalt for Nanoscale Devices," International Conference on Nano Science and Nano Technology 2009, Mokpo, Republic of Korea, November 5 - 6, 2009
51. "Atomic Layer Deposition for Nanoscale Contact Applications", 25th International VLSI/ULSI Multilevel Interconnection Conference, Fremont, CA, USA, October 28 - 30, 2008
52. "Synchrotron X-ray Scattering Study on the Initial Growth of Atomic Layer Deposition Thin Films for the Next Generation MOSFET," Electroceramics XI, Manchester, UK, August 31 - September 3, 2008

Go to the 1st page

Invited Lectures

1. “AI-driven Digital Twin of ALD Process Environment: From Material Synthesis to Process Prediction,” AI + Ab Initio Modeling Forum, Applied Materials, online, August 21, 2026
2. “The One Who Directs AI, The One Who Thinks With AI: From Tool to Thinking Partner — How Asking Becomes a Skill,” VPP Seminar, Future Technology Research Institute, SK hynix, Icheon, Republic of Korea, August 19, 2026
3. “Teaching Atoms to Choose: Fundamentals and Applications of Area-Selective Atomic Layer Deposition,” Graduate School of Semiconductor Materials and Devices Engineering, UNIST, Ulsan, Republic of Korea, May 13, 2026
4. “Building the Future Angstrom by Angstrom: Atomic Layer Deposition Strategies for 3D Nanofabrication in the AI Era,” School of Chemical and Biological Engineering, Zhejiang University, Hangzhou, China, May 6, 2026
5. “A Experimental Learning Computational Approach: From Materials Design to Prediction of Molecular Behavior,” SK hynix, Icheon, Republic of Korea, April 8, 2026
6. “New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication,” Korea-Japan Joint Workshop for ALD, Kumamoto, Japan, February 24, 2026
7. “From Area-Selective Deposition to AI-Designed Materials: New Frontiers in Atomic Layer Deposition,” Korean Institute of Science and Technology, Seoul, Republic of Korea, February 12, 2026
8. “Atomic Layer Innovations: New Pathways for Future Semiconductor Fabrication,” Thin Film Society Webinar Series, Singapore, December 17, 2025
9. “New Paradigms in Atomic Layer Deposition for 3D Semiconductor Device Fabrication,” Department of Mechanical Engineering, Republic of Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea, December 15, 2025
10. “Atomic-Level Solutions for the Scaling Limits of Semiconductors,” SK TriChem, Sejong, Republic of Korea, November 12, 2025
11. “How to Effectively Utilize Large Language Models for Work and Research?” SK hynix, Icheon, Republic of Korea, October 17, 2025
12. “Expanding the Horizons of Atomic Layer Deposition for Next-Generation Semiconductor Fabrication,” TSMC, Taiwan, August 29, 2025

13. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," Department of Materials Science and Engineering, National Taiwan University, Taiwan, August 28, 2025
14. "Ending the 2nd Technological Generation of ASD & Moving Forward to the 3rd for High-Volume Manufacturing," Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, June 27, 2025
15. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," TEL Korea, Suwon, Republic of Korea, June 21, 2025
16. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," Department of Chemical Engineering, Wuhan University of Technology, Wuhan, China, May 16, 2025
17. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," Department of Mechanical Engineering, Huazhong University of Science and Technology, Wuhan, China, May 16, 2025
18. "Cracking Semiconductor Process Technology in One Hour (반도체공정기술1시간만에뽀개기)," Incheon POSCO High School, Incheon, Republic of Korea, May 13, 2025
19. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," Micron Technology, Boise, ID, USA, April 11, 2025
20. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," Intermolecular, Santa Clara, CA, USA, November 22, 2024
21. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," Department of Chemical Engineering, Stanford University, Stanford, CA, USA, November 20, 2024
22. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," Department of Chemical Engineering, Zhejiang University, Hangzhou, China, October 17, 2024
23. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," Department of Mechanical Engineering, POSTECH, Pohang, Republic of Korea, September 19, 2024
24. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," Department of Chemistry, University of Colorado Boulder, Boulder, CO, USA, August 20, 2024

25. "Supercritical Fluid Deposition for Semiconductor Device Fabrications," Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, July 6, 2024
26. "Rediscovery of Atomic Layer Deposition to Overcome the Limitations of Semiconductor Manufacturing," Applied Materials, Santa Clara, CA, USA, May 17, 2024
27. "The 4th Toolbox for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition," Lawrence Livermore National Laboratory, Livermore, CA, USA, November 9, 2023
28. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," Department of Materials Science and Engineering, UNIST, Ulsan, Republic of Korea, October 25, 2023
29. "The Industrial Ecosystem of Si Chips and Atomic Layer Deposition as a Key Nanofabrication Technology," ACS Science Talk, Online, September 8, 2023
30. "Why is My Advisor Always Upset for My Paper Figures?" Department of Materials Science and Engineering, Phenikaa University, Ha Noi, Vietnam, August 11, 2023
31. "The 4th Tool Box for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition," Department of Chemistry, University of Cologne, Cologne, Germany, June 3, 2023
32. "Area Selective Deposition; The 4th Toolbox for Si Device Nanofabrication," Department of Materials Science and Engineering, KAIST, Deajeon, Republic of Korea, May 9, 2023
33. "The 4th Toolbox for Semiconductor Device Nanofabrication beyond ALD," Department of Applied Physics, Kyunghee University, Suwon, Republic of Korea, April 25, 2023
34. "The 4th Toolbox for Semiconductor Device Nanofabrication beyond ALD," Argonne National Laboratory, Chicago, IL, USA, March 29, 2023
35. "The 4th Toolbox for Nanofabrication of Si Devices," Department of Materials Science and Engineering, Phenikaa University, Ha Noi, Vietnam, February 23, 2023
36. "Seam-less Deposition through Gradient Selective Deposition with Homometallic Precursor Inhibitors," Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, February 17, 2023
37. "Improvement of Selectivity in Area Selective Atomic Layer Deposition using a Size Cutter," SK Hynix, Icheon, Republic of Korea, February 16, 2023
38. "Area Selective Deposition of Ru; A Key Process for Si Device Fabrication," Tanaka Precious Metals, Tsukuba, Japan, January 31, 2023
39. "Icephobic Coating through Self-formed Superhydrophobic Layer," Department of Materials Science and Engineering, January 5, 2023

40. "Area Selective Atomic Layer Deposition; The 4th Toolbox for Semiconductor Device Fabrication," Department of Electrical and Electronic Engineering, Sungkyunkwan University, Suwon, Republic of Korea, January 3, 2023
41. "Area Selective Deposition; Past, Present, and Future," Semiconductor Laboratory, Samsung Electronics, Suwon, December 1, 2022
42. "The 4th Tool Box for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition," Department of Chemistry, Carleton University, Ottawa, Canada, September 19, 2022
43. "The 4th Tool Box for Semiconductor Device Nanofabrication beyond Atomic Layer Deposition," Department of Materials Science and Engineering, Hanyang University, Seoul, Republic of Korea, August 18, 2022
44. "How We Evaluate Scientific Papers?" SK hynix, Icheon, Republic of Korea, May 20, 2022
45. "Atomic Layer Deposition for Nanofabrication beyond Deposition," Wuhan University of Technology, online, March 22, 2022
46. "The 4th Tool Box for Nanofabrication beyond ALD; Area Selective Deposition," Applied Materials, online, December 9, 2021
47. "Recent Advances in ALD/ALE Technologies for Next-Generation Semiconductor Applications," KETI, Sungnam, Republic of Korea, October 21, 2021
48. "The 4th Tool Box for Nanofabrication beyond Atomic Layer Deposition; Area Selective Deposition," SK hynix, Icheon, Republic of Korea, October 19, 2021
49. "Application of Area Selective Deposition & Atomic Layer Modulation," SK hynix, Icheon, Republic of Korea, October 1, 2021
50. "Area Selective Deposition And Applications," Department of Materials Science and Engineering, Yeoungnam University, Gyeongsan, Republic of Korea, September 28, 2021
51. "Beyond ALD; Atomic Layer Modulation," SK hynix, Icheon, Republic of Korea, August 24, 2021
52. "Beyond ALD; Atomic Layer Modulation," SAIT, Samsung Electronics, Suwon, Republic of Korea, August 20, 2021
53. "How Can We Prepare Figures for Paper Writing?" Department of Materials Science and Engineering, Pusan National University, Pusan, Republic of Korea, July 12, 2021
54. "Fundamentals of Area Selective Deposition," Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, June 22, 2021
55. "Fundamentals of Area Selective Deposition," SAIT, Samsung Electronics, Suwon, Republic of Korea, June 11, 2021

56. "The Era of Nanofabrication is Coming," Department of Materials Science and Engineering, Seoul University of Science and Technology, Seoul, May 12, 2021
57. "Fundamentals of Area Selective Deposition," SK hynix, Icheon, Republic of Korea, April 26, 2021
58. "Why is My Advisor Always Upset for My Paper Figures?" Department of Materials Science and Engineering, Yeoungnam University, Gyeongsan, Republic of Korea, April 19, 2021
59. "How Can We Prepare Figures for Paper Writing?" Department of Materials Science and Engineering, Hanyang University ERICA, Ansan, Republic of Korea, March 28, 2021
60. "The Era of Atomic Crafting: Area Selective Atomic Layer Deposition," SK Materials, online, February 4, 2021
61. "Beyond ALD; Area Selective Deposition & Atomic Layer Modulation," Department of Chemistry, Charleton University, online, December 15, 2020
62. "The Era of Atomic Crafting," KETI, Sungnam, Republic of Korea, July 21, 2020
63. "Atomic Layer Deposition on Graphene," Semiconductor Laboratory, Samsung Electronics, Suwon, Republic of Korea, July 13, 2020
64. "Science against Pseudoscience," Department of Materials Science and Engineering, Hanyang University ERICA, Ansan, Republic of Korea, November 27, 2019
65. "The Era of Atomic Crafting; Area Selective Atomic Layer Deposition," Department of Chemistry, National University of Singapore, Singapore, September 12, 2019
66. "Science against Pseudoscience," Department of Materials Science and Engineering, Pusan National University, Pusan, Republic of Korea, September 5, 2019
67. "Surface-reactivity-determined Patterning Technology; The Era of Atomic Crafting," Sungkyunkwan University, Suwon, Republic of Korea, August 12, 2019
68. "Surface-reactivity-determined Patterning Technology; The Era of Atomic Crafting," University of California Merced, Merced, CA, USA, July 26, 2019
69. "Science against Pseudoscience," Yeongnam University, Kyeongsan, Republic of Korea, July 17, 2019
70. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, June 26, 2019
71. "Science against Pseudoscience," Myungduk Girls' Highschool, Seoul, Republic of Korea, March 22, 2019
72. "Atomic Layer Deposition; A Route to Atomic Craft," Pusan National University, Pusan, Republic of Korea, November 20, 2018

73. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, November 8, 2018
74. "Surface Energy Tuning in Atomic Scale for Hydrophobic Coating," Department of Chemical Engineering, Hongik University, Seoul, November 11, 2018
75. "Atomic Layer Deposition; A Route to Atomic Craft," Department of Mechanical Engineering, University of Hawaii Manoa, Honolulu, September 24, 2018
76. "A Surface-reactivity-determined Patterning Technology using Atomic Layer Deposition," SK Hynix, Icheon, Republic of Korea, September 6, 2018
77. "Atomic Layer Deposition; A Route to Atomic Craft," Instituto Tecnológico de Aeronáutica, São José dos Campos, Brazil, August 14, 2018
78. "From Surface Property Tuning to Real Applications," Korea Institute of Science and Technology, Seoul, Republic of Korea, July 19, 2018
79. "Atomic Layer Deposition & High-k Dielectrics for Emerging Devices," Samsung Advanced Institute of Technology, Suwon, Republic of Korea, July 11, 2018
80. "Thin Film Growth & Area Selective Atomic Layer Deposition," Semiconductor Research Laboratory, Samsung Electronics, Suwon, Republic of Korea, June 18, 2018
81. "Thin Film Growth & Atomic Layer Deposition," Samsung Electronics, Suwon, Republic of Korea, June 14, 2018
82. "Area-selective Atomic Layer Deposition; A Surface-reactivity-determined Patterning Technology," Applied Materials, Santa Clara, CA, USA, June 7, 2018
83. "How Goals Can Make Our Life Better?" Myungduk Girls' Highschool, Seoul, Republic of Korea, May 26, 2018
84. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 3, 2018
85. "Atomic Layer Deposition; A Route to Atomic Craft," Materials Chemistry Laboratory, Institute for Basic Science, Daejeon, Republic of Korea, May 1, 2018
86. "Hydrophobic Coating by High Tech and Low Tech Approaches," Department of Materials Science and Engineering, Yeongnam University, Kyeongsan, Republic of Korea, December 19, 2017
87. "From Surface Property Tuning to Real Applications," Department of Materials Science and Engineering, Kyung Hee University, Suwon, Republic of Korea, November 11, 2017

88. "Atomic Layer Deposition, A Key Player in Semiconductor Technology," Lam Research, Fremont, CA, USA, July 20, 2017
89. "From Surface Property Tuning to Real Applications," Department of Nano and Energy Engineering, Pusan National University, Pusan, Republic of Korea, June 16, 2017
90. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 17, 2017
91. "How Nanotechnologies Can Make Our Life Better?," Myungduk Highschool, Seoul, Republic of Korea, April 21, 2017
92. "Electronic Textiles fabricated by Atomic Layer Deposition for Wearable Electronics," Korea Institute of Industrial Technology (KITECH), Ansan, Republic of Korea, February 28, 2017
93. "Electronic Textiles Fabricated by Atomic Layer Deposition for Wearable Electronics and Bio Applications," Department of Chemical Engineering, Chungnam National University, Daejeon, Republic of Korea, February 23, 2017
94. "Surface Functionalization of Graphene by Atomic Layer Deposition," Electronic Convergence Material and Device Research Center, Republic of Korea Electronics Technology Institute (KETI), Sungnam, Republic of Korea, December 5, 2016
95. "Atomic Layer Deposition Beyond High Technology; Electronic Textiles, Hydrophobic Filter, and Transparent 2D Heater," Electronic Convergence Material and Device Research Center, Republic of Korea Electronics Technology Institute (KETI), Sungnam, Republic of Korea, November 11, 2016
96. "Atomic Layer Deposition Beyond High Technology; Electronic Textiles, Hydrophobic Filter, and Transparent 2D Heater," School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, November 9, 2016
97. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, October 11, 2016
98. "Atomic Layer Deposition; An Essential Tool for Nanofabrication," Department of Chemistry, Thammasat University, Bangkok, Thailand, October 6, 2016
99. "Atomic Layer Deposition; An Essential Tool for Nanofabrication," Department of Materials Science and Engineering, Kasetsart University, Bangkok, Thailand, October 5, 2016
100. "Surface Functionalization for Property Tuning," Division of Materials Science and Engineering, Hanyang University, Seoul, Republic of Korea, September 27, 2016
101. "Conductive Textile for Sensors and Heaters," School of Materials Science and Engineering, Yongnam University, Gyeongsan, Republic of Korea, September 21, 2016

102. "Atomic Layer Deposition; An Essential Tool for Nanofabrication," Tera Semicon, Anseong, Gyeonggi, Republic of Korea, September 9, 2016
103. "Atomic Layer Deposition; An Essential Tool for Nanotechnology," Department of Materials Science and Engineering, Sejong University, Seoul, Republic of Korea, June 16, 2016
104. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 20, 2016
105. "Surface Functionalization for Property Tuning," School of Advanced Materials Engineering, Sungkyunkwan University, Suwon, Republic of Korea, May 12, 2016
106. "Surface Functionalization for Property Tuning," Department of Mechanical Engineering, Kyunghee University, Suwon, Republic of Korea, May 9, 2016
107. "Defect Healing of Graphene," School of Materials Science and Engineering, Yongnam University, Gyeongsan, Republic of Korea, April 1, 2016
108. "Platinum Atomic Layer Deposition," Department of Mechanical Engineering, Pusan National University, Yongin, Republic of Korea, December 29, 2015
109. "Atomic Layer Deposition for Nanofabrication," Myungji University, Yongin, Republic of Korea, November 6, 2015
110. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, October 5, 2015
111. "Atomic Layer Deposition; An Essential Tool for Nanotechnology," Department of Energy Science, Sungkyunkwan University, Suwon, Republic of Korea, September 30, 2015
112. "Tutorial: Atomic Layer Deposition," Advanced Metallization Conference 2015, Seoul, Republic of Korea, September 16, 2015
113. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 8, 2015
114. "Applications of Atomic Layer Deposition Pt," Tanaka Research Center, Tsukuba, Japan, April 2, 2015
115. "Series Class of Atomic Layer Deposition," Samsung Electronics, Hwasung, Republic of Korea, March 5-6, 2015
116. "Research Topics of the Lee's Group," Wonik IPS, Pyeongtaek, Republic of Korea, March 2, 2015
117. "Material Synthesis and Applications by Atomic Layer Deposition," Graduate School of Convergence Science and Technology, Seoul National University, Seoul, Republic of Korea, January 19, 2015

118. "Low Temperature Atomic Layer Deposition of Pt for Electronic Textile Applications," State Key Laboratory, Nanjing Tech University, Nanjing, China, January 16, 2015
119. "Surface Functionalization by Atomic Layer Deposition," Samsung Medical Center, Seoul, Republic of Korea, November 26, 2014
120. "Atomic Layer Deposition," e-Nanoschool 2014 by Korea Nano Technology Research Society, 6 on-line classes from November 18 to December 4, 2014
121. "Atomic Layer Deposition on 2D Materials," School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Ulsan, November 5, 2014
122. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, October 8, 2014
123. "Atomic Layer Deposition of Pt and Applications," Shonan Site, Tanaka Kikinzoku Kogyo K.K., Hiratsuka, Japan, September 19, 2014
124. "Atomic Layer Deposition of Pt/TiO2 for VOC Removal," Head Office, Tanaka Kikinzoku Kogyo K.K., Tokyo, Japan, September 18, 2014
125. "Atomic Layer Deposition of Metals on 2D Materials and Applications," Department of Materials Science and Engineering, Hanyang University, Seoul, Republic of Korea, July 30, 2014
126. "Pt/Carbon by Atomic Layer Deposition for Catalyst, Electrode, and Sensor Applications," Electronic Materials Research Center, Republic of Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea, July 28, 2014
127. "Atomic Layer Deposition; from Fundamentals to Applications," Semiconductor Equipment Technology Center, Chunan, Republic of Korea, May 15, 2014
128. "Atomic Layer Deposition for Emerging Applications," Department of Polymer Science and Engineering, Korean National University of Transportation, Chungju, Republic of Korea, December 23, 2013
129. "Atomic Layer Deposition for Interconnect Technology," Inter-University Semiconductor Research Center, Seoul National University, Seoul, Republic of Korea, November 29, 2013
130. "What is a Scientific Paper?," School of Materials Science and Engineering, Pusan National University, Pusan, Republic of Korea, November 7, 2013
131. "Atomic Layer Deposition of Pt on Carbon Surfaces; Graphite and Graphene," Korea Institute of Machinery and Materials, Daejun, Republic of Korea, September 26, 2013
132. "Atomic Layer Deposition of Pt on Carbon Surfaces; Graphite and Graphene," Korea Research Institute of Standards and Science, Daejun, Republic of Korea, September 26, 2013

133. "Atomic Layer Deposition of Pt on Carbon Surfaces," Department of Mechanical Engineering, Republic of Korea University, Seoul, Republic of Korea, August 20, 2013
134. "Atomic Layer Deposition of Pt on Carbon Surfaces" Department of Materials Science and Engineering, Yeongnam University, Daegu, Republic of Korea, June 26, 2013
135. "Atomic Layer Deposition of Pt on Carbon Surfaces," Graduate School of Energy, Environment, Water, and Sustainability, Republic of Korea Advanced Institute of Science and Technology, Daejeon, Republic of Korea, May 8, 2013
136. "Atomic Layer Deposition for Nanotechnologies," Department of Chemical Engineering, Inha University, Incheon, Republic of Korea, May 3, 2013
137. "Nucleation & Growth of ALD Metal on Carbon Surfaces," Korean ALD Workshop 2013, Seoul, Republic of Korea, April 26, 2013
138. "How Nanotechnologies Can Make Our Life Better?," Myungduk Highschool, Seoul, Republic of Korea, April 23, 2013

Go to the 1st page

Professional Activities

CURRENT APPOINTMENTS

- Executive Editor of Chemistry of Materials, ACS Publications, 2025 - present
- Visiting Consulting Professor, SK hynix, 2026 - present
- Program Committee, ALD/ALE, 2018 - present
- Steering Committee, ALD, 2026 - present
- Program Committee, AVS, 2025 - present
- Program Committee, Euro CVD, 2025 - present
- International Committee, Asian Pacific ALD, 2018 - present
- Editorial Board Member, 2015 - present, Electronic Materials Letters (http://link.springer.com/journal/13391)

PREVIOUS APPOINTMENTS

- Associate Editor of Chemistry of Materials, ACS Publications, 2018 - 2025
- Chair, International Conference ALD/ALE 2025, 2025
- Visiting Consulting Professor, SK hynix, 2021-2022
- Chair, Workshop on Area Selective Deposition, 2023
- Co-organizer, ACS Award in Surface Chemistry: Symposium in honor of Stacey F. Bent, 255th ACS National Meeting, 2018, New Orleans, LA
- Scientific Program Committee, 7th International Conference on Microelectronics and Plasma Technology (ICMAP 2018), 2018, Republic of Korea
- Guest Editor for ACS Virtual Issue, "Recent Advances in Atomic Layer Deposition," 2016 (http://pubs.acs.org/page/vi/advances-ald.html)
- Organizing committee, Advanced Metallization Conference 2015 (ADMETA 2015), 2015, Republic of Korea
- General Secretary, Devision of Materials Science Convergence in The Korean Institute of Metals and Materials, 2015, Republic of Korea
- Organizer, 2nd Korean Meeting on Atomic Layer Deposition for Emerging Technologies, supported by The Korean Information Display Society, 2014, Republic of Korea
- Advisory Committee, Devision of Nanoprocessing & Nanoanalysis Tools, Selection of Nano R&BD Topics, 2014, Republic of Korea
- Symposium Organizer, Division of Equipments and Tools for Display Technology, The 14th International Meeting on Information Display, 2014, Republic of Korea

- Symposium Organizer and Chair, “Atomic Layer Deposition for Energy Applications,” Korean Ceramic Society Fall Meeting 2013, 2013, Republic of Korea
- Organizer, 1st Korean Meeting on Atomic Layer Deposition for Emerging Technologies, supported by The Korean Information Display Society, 2013, Republic of Korea
- Advisory Committee, Devision of Nanoprocessing & Nanoanalysis Tools, Nano Roadmap 2nd Edition, 2013, Republic of Korea

Go to the 1st page

Fellowship and Honor

- Reseach and Science Award of Incheon Metropolitan City 2017
  - Date: 4/9/2018
- Young Scientist Award, The Korean Institute of Metals and Materials 2015
  - Date: 4/23/2015
- Best Paper Award, Material Research Society of Korea 2009
  - Title: High quality nickel atomic layer deposition for nanoscale contact applications
  - Date: 5/21/2009
- Graduate Student Award (Outstanding Paper), Dept. of MSE, POSTECH
  - Title: Spontaneous formation of vertical magnetic metal nanorod arrays during plasma enhanced atomic layer deposition
  - Date: 12/15/2008
- Samsung Award (Best Paper Award), The 15th Korean Conference on Semiconductor
  - Title: High quality epitaxial $CoSi_2$ using plasma nitridation-mediated epitaxy
  - Date: 2/22/2008
- Graduate Student Award (Outstanding Conf. Presentation), Dept. of MSE, POSTECH
  - Title: High quality epitaxial $CoSi_2$ using plasma nitridation-mediated epitaxy
  - Date: 12/15/2007
- Best Poster Award, Material Research Society 2007 Fall Meeting
  - Title: Analytical study on initial growth stage of metal atomic layer deposition by synchrotron radiation X-ray reflectivity analysis
  - Data: 11/28/2007
- University fellowship, Sungkyunkwan University
  - Period: 2002 - 2004

Go to the 1st page

Teaching Activities

- MSE Laboratory 1
  - For sophomore, spring
- Modern Physics
  - For sophomore, fall
- Semiconductori Device Nanofabrications 1
  - For junior, spring
- Semiconductori Device Nanofabrications 2
  - For junior, fall
- Materials Analysis
  - For graduate student
- Materials Kinetics
  - For graduate student
- Thin Film Deposition
  - For graduate student

Go to the 1st page

University and Departmental Committees

- Advisory Committee, INU Center for Research Facilities, 2014 – 2019
- Committee, Task Force Team of College of Engineering, INU, 2015 – 2019
- Committee, Graduate School of INU, 2018 – 2019
- Committee, Global Research Track, INU, 2019 – 2020
- Department Chair, Department of Materials Science and Engineering, INU, 2022 – 2023

Press

1. “Electronic Textiles fabricated by Low Temperature Atomic Layer Deposition,” in The Kukmin Daily, Yonhap News, Kyeonggi Daily, Herald Economy, NSP News Agency, Maeil Daily, Daehak News, Bridge News,12/5/2016.
2. “Defect Healing of Graphene by Galvanic Displacement Reaction for Transparent Electrode,” in YP News, The Electronic Times, Veritas Alpha, Energy & Economy Times, 4/20/2016.
3. “Guest Editor for ACS Virtual Issue (Recent Advances in Atomic Layer Deposition),” in Aju Business Daily, Gukje News, Sport Kyunghyang, Bridge News, Asia Today, Kyeonggi Daily, Kyeongin Daily, 4/14/2016.
4. "INU Department of Materials Science Engineering and Samsung Advanced Institute of Technology Publish Nano-Process Research in Nature Communications," in Daihan Economy, Sports Kyunghyang, eNews Today, Kiho Ilbo, 12/13/2022. [Link]
5. "Prof. Han-Bo-Ram Lee (INU) and Prof. Il-Kwon Oh (Ajou University) Transfer KRW 1.175 Billion in Semiconductor Research Equipment Technology," in Daihan Economy, Seoul Ilbo, Gyeonggi Domin Ilbo, INU News, 12/20/2023. [Link]
6. "INU Team Publishes Ultra-Thin Insulating Film Technology for Next-Generation Semiconductors in ACS Nano," in Naewoeilbo, Jeonmae, Asia Ilbo, 2025. [Link]
7. "INU Research Team's ALD Equipment and Process Technology Transferred to CN1, Opening a New Horizon in AI Computing Speed," in Naewoeilbo, Bridge Economy (Viva100), INU News, 9/25/2025. [Link 1] [Link 2]
8. "INU Prof. Han-Bo-Ram Lee Research Team Publishes Next-Generation Semiconductor Ruthenium Interconnect Technology in JACS," in Gyeongin Newspaper, Gukje News, Newstown, Gyeonginmaeil, Korean Media News, Newsis, Golden Times, 6/10/2026. [Link]

Go to the 1st page